

2D Yukawa Model applied to Hamiltonian Truncation Effective Theory

Master Thesis

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Abstract

The thesis demonstrates the implementation of the 2D Yukawa potential in equal time Hamiltonian truncation. The theory couples scalars and spinors in a three point interaction and is UV divergent. It is shown, how this property is treated using the renormalization concept of Delouche et al.[2] and the description of an effective Hamiltonian, given by Cohen et al.[1].

Thereby, a non-perturbative, numerical method, valid for any value of the coupling is presented.

The free Hamiltonian is used to build up the Fock basis, where momentum conservation and charge conservation is utilized. Additionally the implementation of charge conjugation symmetry is described.

The diagonalization of the Hamiltonian yields the energy spectrum, for varying couplings and energy. It became apparent that C-odd states can have negative energy with respect to the C-even vacuum.

Moreover, a variation of the effective Hamiltonian is explored. The local approximation is dropped, leaving an expansion in the coupling only. In doing so, the power counting arguments of [1] are generalized and computational difficulties are solved.

The new method showed to be less suitable for extracting the low energy spectrum, but further applications are discussed.

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1 Introduction

In modern physics, there are a large number of possible QFTs. In particular for strong coupling, the conventional perturbation methods are not sufficient to extract observables. Thus, new tools need to be devised to explore the theories.

Especially of interest are lattice methods[19], conformal bootstrap[18] and Hamiltonian Truncation. The theses focus on the latter one.

Hamiltonian Truncation offers a non-perturbative model, suitable for any kind of Hamiltonian. There are no requirements for symmetries of the theory. Instead, preserved quantities can be used to split the Fock space and accelerate the calculations. The only restriction is, that the theory is small in the UV, i.e. the coupling must be relevant. Nevertheless, it provides a tool for theories, which might be beyond boundaries of other methods.

The idea of the method[1] is to split the Hamiltonian in a free part and an interaction, as

$$H = H_0 + V. \quad (1)$$

H_0 is the part, which can be diagonalized exactly and is used to build up the Fock basis. To discretize the free energy spectrum

$$H_0|i\rangle = E_i|i\rangle, \quad (2)$$

the theory is defined on a finite volume associated with a parameter R .

Additionally, an energy cutoff E_{max} is chosen. By taking only states with $E_i < E_{max}$ into account the Fock space becomes finite. This way the low energy regime is separated.

The next step is to diagonalize the Hamiltonian, including the interaction. This provides a basis change from the free particle basis to diagonalized mixed basis. The eigenvalues give then the energy spectrum for stationary states.

By varying $R \rightarrow \infty$ and $E_{max} \rightarrow \infty$, the convergence of the spectrum lead to observables of the full theory. In doing so, the method is based on variational methods[8], in particular the Rayleigh-Ritz method.

The disadvantage of truncation is, that the number of states grow exponentially with E_{max} (shown in section 13). To improve the convergence[17] for finite E_{max} an effective Hamiltonian formulation[16] can be applied.

The idea is to add corrections from outside the Hilbert space. It can be thought of as "integrating out" higher energy states. This is done by matching an physical observable in the infinite Fock space with the effective Hamiltonian in the truncated space. Since E_{max} introduces a separation of scales, this is done in a $1/E_{max}$ expansion[1]. Improving the convergence is then a matter of systematically calculating the effective Hamiltonian up to the order, where the desired precision is reached.

Here in particular of interest is the Gross-Neveu model

$$\mathcal{L} = \sqrt{-g} \left[\bar{\psi}_a (i\not{\partial} - m) \psi_a + \frac{g^2}{2N} (\bar{\psi}_a \psi_a)^2 \right], \quad (3)$$

which couples N types of fermion fields ψ_a in a four point interaction with coupling strength g . It shares the property of asymptotic freedom with QCD. Therefore, it might be used as a toy model [4]. Initially the fermions are taken massless, but that changes due to symmetry breaking similar to the Higgs mechanism [4][6].

In this work the theory is studied after the symmetry breaking by adding the fermion mass by hand. Moreover the calculations are done in two dimensions and one type of fermions are considered, i.e. $N = 1$.

In 2D, the coupling g is marginal, which makes the truncation method difficult. The problem can be avoided by approximating the fermion interaction by a Yukawa interaction with a massive scalar field. Introducing the field ϕ with mass M gives

$$\mathcal{L} = \sqrt{-g} [\bar{\psi}(i\not{D} - m)\psi + \frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}M^2\phi^2 - g\phi\bar{\psi}\psi]. \quad (4)$$

If $M \gg m$ is big enough, this model converges to the Gross-Neveu model.

The theses is focused on applying the truncation method to the Yukawa interaction, which is the first step of this process. The goal is to provide a detailed description, how the methods of truncation are utilized.

For this purpose, sections 2-5 explain the quantization, 6-8 show explicitly, how symmetries are used to build up the Fock basis and section 9 demonstrate the implementation. In the latter one, also a computational treatment of duplets are shown. Those appear, because charge conjugation symmetry introduce a basis change, where some charge conjugated states are build up by two of the initial states.

Moreover, Yukawa theory is UV divergent. The treatment of such interactions in truncation will be tested, following the description of Delouche et al.[2]. Together with the effective Hamiltonian, this will be derived in sections 10 and 11. For the effective Hamiltonian the method of [1] is demonstrated at leading order of the $1/E_{max}$ -expansion.

As a new method, it will be explored, if the local approximation can be dropped, leaving an expansion in the coupling only. This is of interest for future work, because this new technique might be more suitable for the limit to the Gross-Neveu model with scalar mass $M \rightarrow \infty$. A discussion of this matter is presented in section 12.

Lastly in 14 the results are compared to the work of Brooks and Frautschi[3].

2 Yukawa Model

2.1 Preface: Metric Transformation

For a generic metric, the action is formally written as

$$S = \int_{\mathcal{M}} \mathcal{L}(x) d\mu_g, \quad (5)$$

where the integral runs over the Manifold $\mathcal{M}$ with the measure $d\mu_g$. μ_g is defined, using that the action should not depend on a specific choice of coordinates, therefore the integral has to be invariant under a coordinate change $x^\mu \rightarrow \tilde{x}^\mu$. This can be accomplished by defining

$$S = \int_{\mathcal{M}} \mathcal{L}(x) d\mu_g \equiv \int \mathcal{L}(x) \sqrt{|g(x)|} d^d x, \quad (6)$$

where $g(x)$ is the metric. The derivatives, coming from the metric transformation, can be factored out. Therefore,

$$\sqrt{|g(x)|} d^d x = \sqrt{|g(x)|} \left| \det \left(\frac{\partial x^\mu}{\partial \tilde{x}^\nu} \right) \right| d^d \tilde{x} = \sqrt{|\tilde{g}(\tilde{x})|} d^d \tilde{x}, \quad (7)$$

and the expression eq.(6) is indeed coordinate independent[9].

2.2 Coordinate Change to Cylindrical Coordinates

The transformation from Cartesian coordinates of Lorentzian signature to polar coordinates is

$$\begin{pmatrix} x \\ iy \end{pmatrix} = \begin{pmatrix} r \cos(\varphi) \\ r \sin(\varphi) \end{pmatrix}, \quad ds^2 = dx^2 - dy^2 \quad (8)$$

$$\Rightarrow \begin{pmatrix} r \\ \varphi \end{pmatrix} = \begin{pmatrix} \sqrt{x^2 - y^2} \\ \arctan(i \frac{y}{x}) \end{pmatrix}, \quad ds^2 = dr^2 - r^2 d\varphi^2. \quad (9)$$

φ is an angle and will be related to the coordinate, which goes around the cylinder. Therefore, the Lagrangian would need to be 2π -periodic (one could also choose an 2π -anti-periodic condition for fermions).

The further transformation $r = e^\xi$, brings the metric closer to the shape of a cylinder. As expected for a time variable, $\xi \in \{-\infty, \infty\}$, for $r \in \{0, \infty\}$. Using $\frac{dr}{d\xi} = r$, the metric becomes

$$ds^2 = r^2(d\xi^2 - d\varphi^2). \quad (10)$$

This is not a cylinder yet, since the radius $r = e^\xi$ is coordinate dependent. It is possible to change that with a Weyl-transformation[9], such that the metric get stretched by a coordinate dependent factor $(R/r)^2$. The coordinate transformation yields

$$g'^{\mu'\nu'} = \left(\frac{R^2}{r(\xi)^2} \right) g^{\mu\nu} \Leftrightarrow ds^2 = R^2(d\xi^2 - d\varphi^2) \quad (11)$$

$$\rightarrow ds^2 = d\tau^2 - d\theta^2 \quad (12)$$

The last metric eq.(12) will be used henceforth. It is the metric of a cylinder with radius R .

In this coordinates the action takes the form (as derived in eq.(6))

$$S = \int \sqrt{|g(\tau, \theta)|} d\tau d\theta \mathcal{L}(\tau, \theta) = \int d\tau d\theta \mathcal{L}(\tau, \theta). \quad (13)$$

The Lagrangian $\mathcal{L}(\tau, \theta)$ in the new coordinates will be derived analogous to the derivation in Cartesian coordinates, but θ is $2\pi R$ periodic now. Other curved space effects, like a change of the derivative are neglected, because the interest lies in the limit $R \rightarrow \infty$. Then, the cylindrical metric eq.(12) matches the Cartesian metric.

So, the action of the Yukawa potential is

$$\begin{aligned} S &= \int d\tau d\theta \left(\mathcal{L}_\phi^{free} + \mathcal{L}_\psi^{free} + \mathcal{L}^{int} \right) \\ &= \frac{1}{2}(\partial\phi)^2 - \frac{1}{2}M^2\phi^2 + \bar{\psi}(i\partial - m)\psi + g\phi\bar{\psi}\psi \end{aligned} \quad (14)$$

2.3 Dirac Representation

In an infinitesimal surrounding cylindrical space is flat, thus the Lie Algebra[10][11] still upholds. This justifies the use of the same γ matrices as in flat space. Here in particular, the Dirac matrices

$$\gamma^0 = \begin{pmatrix} & -i \\ i & \end{pmatrix}, \quad \gamma^1 = \begin{pmatrix} & i \\ i & \end{pmatrix} \quad \Leftrightarrow \{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu} = 2 \begin{pmatrix} 1 & \\ & -1 \end{pmatrix} \quad (15)$$

are chosen to be imaginary¹. The fermion part of the Lagrangian eq.(15) can be written as

$$\mathcal{L} = i \begin{pmatrix} \psi_1^\dagger & \psi_2^\dagger \end{pmatrix} \begin{bmatrix} & -1 \\ 1 & \end{bmatrix} \left(\begin{bmatrix} & \partial_\tau - \partial_\theta \\ -(\partial_\tau + \partial_\theta) & \end{bmatrix} - m \right) \begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix} \quad (16)$$

$$\Leftrightarrow \mathcal{L} = i\psi_1^\dagger \partial_+ \psi_1 + i\psi_2^\dagger \partial_- \psi_2 + im(\psi_1^\dagger \psi_2 - \psi_2^\dagger \psi_1), \quad (17)$$

where the abbreviation

$$\partial_+ \equiv \partial_\tau + \partial_\theta \quad , \quad \partial_- \equiv \partial_\tau - \partial_\theta. \quad (18)$$

is used.

3 Boson Quantization

The following chapters are based on standard quantization procedures, which can be found for example in [12][13].

3.1 Equation of Motion

The particle associated with the Lagrangian eq.(15)

$$\mathcal{L}_\phi^{free} = \frac{1}{2}(\partial_\tau \phi)^2 - \frac{1}{2}(\partial_\theta \phi)^2 - \frac{1}{2}M^2 \phi^2$$

satisfies the Euler-Lagrange equation

$$\square \phi = -M^2, \quad (19)$$

yielding the equation of motion (e.o.m.). The solution is a scalar field associated with bosons. Therefore, the periodic boundary condition $\phi(\theta + 2\pi R) = \phi(\theta)$ must be satisfied. The generic solution is a superposition of functions in the form $f_k e^{ik\theta/R}$, with $k \in \mathbb{Z}$. That leads to the ansatz

$$\phi = \sum_{k=-\infty}^{\infty} \frac{N_k}{\sqrt{2\pi R}} \left(a_k e^{-i(\Omega_k \tau - k\theta/R)} + a_k^\dagger e^{i(\Omega_k \tau - k\theta/R)} \right). \quad (20)$$

The factor $1/\sqrt{2\pi R} = 1/\sqrt{V}$ is a conventional prefactor of the sum related to the finite volume of the θ -coordinate. The other variables will be determined and interpreted through the chapter.

The e.o.m yields

$$\begin{aligned} \square \phi &= \left[-\Omega_k^2 + \left(\frac{k}{R} \right)^2 \right] \phi = -M^2 \phi \\ \Rightarrow \Omega_k &= \sqrt{M^2 + \left(\frac{k}{R} \right)^2}, \end{aligned} \quad (21)$$

Where Ω is conventionally chosen to be positive. Otherwise the interpretation of a and $a^\dagger$ would switch.

¹The reasons for choosing these γ 's are:

- off-diagonal: In the kinetic part the two field components split in to different terms
- imaginary: The field of the antiparticle will have the same mathematical structure as the particle (which shortens the equations)

3.2 Commutation Relations

The canonical commutation relation is given as

$$[\phi(\tau, \theta_x), \pi(\tau, \theta_y)] = i\delta(\theta_x - \theta_y); \quad [\phi, \phi] = [\pi, \pi] = 0, \quad (22)$$

where π is defined as

$$\pi = \Pi^0 \equiv \frac{\partial \mathcal{L}}{\partial(\partial_\tau \phi)} = \sum_k \frac{N_k}{\sqrt{2\pi R}} i\Omega_k \left(-a_k e^{-i(\Omega_k \tau - k\theta/R)} + a_k^\dagger e^{i(\Omega_k \tau - k\theta/R)} \right). \quad (23)$$

Using Fourier transformation

$$\int_0^{2\pi R} \frac{d\theta}{2\pi R} e^{in\theta/R} = \delta_{n,0}, \quad (24)$$

The commutation relations in Fourier space are

$$\begin{aligned} a_k &= \int d\theta_x \frac{1}{\sqrt{2\pi R} N_k} e^{i(\Omega_k \tau - k\theta_x/R)} \frac{1}{2} [\phi(\theta_x) - \frac{1}{i\Omega_k} \pi(\theta_x)] \\ a_l^\dagger &= \int d\theta_y \frac{1}{\sqrt{2\pi R} N_l} e^{-i(\Omega_l \tau - l\theta_y/R)} \frac{1}{2} [\phi(\theta_y) + \frac{1}{i\Omega_l} \pi(\theta_y)] \end{aligned} \quad (25)$$

$$\begin{aligned} \Leftrightarrow [a_k, a_l^\dagger] &= \int \frac{d\theta_x d\theta_y}{2\pi R N_k N_l} e^{i(\Omega_k - \Omega_l)\tau} e^{-i(k\theta_x - l\theta_y)/R} \left(\frac{1}{4i\Omega_k} [\phi(\theta_x), \pi(\theta_y)] - \frac{1}{4i\Omega_l} [\pi(\theta_x), \phi(\theta_y)] \right) \\ \Leftrightarrow [a_k, a_l^\dagger] &= \frac{\delta_{kl}}{2\Omega_k N_k^2} = \delta_{k,n}, \quad \text{with } N_k = \frac{1}{\sqrt{2\Omega_k}}. \end{aligned} \quad (26)$$

For simplicity states are chosen to be normalized to one, i.e.

$$\langle f|i \rangle = \delta_{fi}, \quad (27)$$

which is directly related to the choice of normalization N_k . A multi particle state $|n_k\rangle$ with n particles of momentum k is taken to be proportional to $(a_k^\dagger)^n |0\rangle$. Using the normalization condition, the operators act as

$$\begin{aligned} a_k^\dagger |n_k\rangle &= \sqrt{n_k + 1} |n_k + 1\rangle \\ a_k |n_k\rangle &= \sqrt{n_k} |n_k - 1\rangle, \quad a_k |0\rangle = 0 \end{aligned} \quad (28)$$

and complex conjugated equations.

Therefore $a^\dagger$ is interpreted as creation operator and a as annihilation operator.

3.3 Energy

The action has to be invariant under a time translation. Following from Noether Theorem[12], the observable

$$:H := \int d\theta : \Pi^0 \partial_0 \phi : - : \mathcal{L} : \quad (29)$$

is conserved. The notation $::$ indicates the usual normal ordering[12].

Taking only the free part $\mathcal{L}_\phi^{free}$ into account, yields the free energy

$$\begin{aligned}
:H_0: &:= \int d\theta \left[\frac{1}{2} : (\partial_\tau \phi)^2 : + \frac{1}{2} : (\partial_\theta \phi)^2 : + \frac{1}{2} M^2 : \phi^2 : \right] \\
\text{with } \begin{cases} \int d\theta \frac{1}{2} : (\partial_\tau \phi)^2 : &= \sum_k \frac{\Omega_k}{2\Omega_k} \left[a_k^\dagger a_k - \frac{1}{2} a_k^\dagger a_{-k}^\dagger e^{2i\Omega_k \tau} - \frac{1}{2} a_k a_{-k} e^{-2i\Omega_k \tau} \right] \\ \int d\theta \frac{1}{2} : (\partial_\theta \phi)^2 : &= \sum_k \frac{(k/R)^2}{2\Omega_k} \left[a_k^\dagger a_k + \frac{1}{2} a_k^\dagger a_{-k}^\dagger e^{2i\Omega_k \tau} + \frac{1}{2} a_k a_{-k} e^{-2i\Omega_k \tau} \right] \\ \int d\theta \frac{1}{2} M^2 : (\phi)^2 : &= \sum_k \frac{M^2}{2\Omega_k} \left[a_k^\dagger a_k + \frac{1}{2} a_k^\dagger a_{-k}^\dagger e^{2i\Omega_k \tau} + \frac{1}{2} a_k a_{-k} e^{-2i\Omega_k \tau} \right] \end{cases} \\
\Leftrightarrow :H_0: &:= \sum_k \frac{1}{2\Omega_k} \left[\left(\Omega_k^2 + \left(\frac{k}{R} \right)^2 + M^2 \right) a_k^\dagger a_k - \left(\Omega_k^2 - \left(\frac{k}{R} \right)^2 - M^2 \right) (a_k^\dagger a_{-k}^\dagger \dots + a_k a_{-k} \dots) \right] \\
\Leftrightarrow :H_0: &:= \sum_k \Omega_k a_k^\dagger a_k.
\end{aligned} \tag{30}$$

$a_k^\dagger a_k$ is a counting operator. Therefore the eigenvalue of $:H_0:$ is indeed the free energy of the boson part.

In summary the particle associated to the free Lagrangian $\mathcal{L}_\phi^{free}$ is a scalar field given by

$$\phi = \sum_k \frac{1}{\sqrt{2\pi R \cdot 2\Omega_k}} \left(a_k e^{-i(\Omega_k \tau - k\theta/R)} + a_k^\dagger e^{i(\Omega_k \tau - k\theta/R)} \right), \text{ with } \Omega_k = \sqrt{M^2 + \left(\frac{k}{R} \right)^2}. \tag{31}$$

4 Fermion Quantization

The aim of this section is to find solutions to the Lagrangian eq.(15)

$$\mathcal{L}_\psi^{free} = \psi^\dagger \gamma^0 (i\gamma^\mu \partial_\mu - m) \psi = i\psi_1^\dagger \partial_+ \psi_1 + i\psi_2^\dagger \partial_- \psi_2 + im(\psi_1^\dagger \psi_2 - \psi_2^\dagger \psi_1)$$

4.1 Equation of Motion

In the e.o.m.(Euler-Lagrange equation) ψ_1 and ψ_2 are coupled, since

$$\begin{aligned}
\partial_+ \psi_1^\dagger &= -m\psi_2^\dagger, & \partial_+ \psi_1 &= -m\psi_2 \\
\partial_- \psi_2^\dagger &= m\psi_1^\dagger, & \partial_- \psi_2 &= m\psi_1.
\end{aligned} \tag{32}$$

Decoupled e.o.m. are gained, using

$$[\partial_\tau^2 - \partial_\theta^2] \psi_1 = \partial_- \partial_+ \psi_1 = -m\partial_- \psi_2 = -m^2 \psi_1 \tag{33}$$

$$\Leftrightarrow \square \psi_2 = -m^2 \psi_2, \tag{34}$$

which are the same as for the boson field.

4.1.1 Decoupled Equations of Motion

Analogous to the scalar field, the solution can be written as a superposition of Fourier factors $e^{ik\theta/R}$ with $k \in \mathbb{Z}$, where periodic boundary conditions are chosen².

²For fermions one could as well choose anti periodic boundary conditions

The decoupled e.o.m. eq.(35) are the same as for the bosons. So, the mathematical structure can be kept and

$$\psi = \sum_k \frac{N_k}{\sqrt{2\pi R}} \left[\mathbf{v}_k c_k e^{-i(\omega_k \tau - k\theta/R)} + \mathbf{u}_k^* d_k^\dagger e^{i(\omega_k \tau - k\theta/R)} \right], \quad (36)$$

with $\omega_k = \sqrt{m^2 + \left(\frac{k}{R}\right)^2}$

The vectors $\mathbf{v}$, $\mathbf{u}$ can be derived using the coupled equations.

4.1.2 Coupled Equation of Motion

The coupled equations eq.(33) are explicitly expressed in the form of

$$\begin{aligned} \partial_+ \psi_{1,k} &= -i\left(\omega_k - \frac{k}{R}\right) v_{1,k} c_k e^{-i(\omega_k \tau - k\theta/R)} + i\left(\omega_k - \frac{k}{R}\right) u_{1,k}^* d_k^\dagger e^{i(\omega_k \tau - k\theta/R)} \\ \stackrel{!}{=} -m \psi_{2,k} &= -m v_{2,k} c_k e^{-i(\omega_k \tau - k\theta/R)} - m u_{2,k}^* d_k^\dagger e^{i(\omega_k \tau - k\theta/R)} \end{aligned} \quad (37)$$

and similar the other equations. m can be rewritten as $m = \sqrt{\omega_k^2 - \left(\frac{k}{R}\right)^2} = \sqrt{\omega_k + \frac{k}{R}} \sqrt{\omega_k - \frac{k}{R}}$. All equations are equivalent to

$$i\sqrt{\omega_k - \frac{k}{R}} v_{1,k} = \sqrt{\omega_k + \frac{k}{R}} v_{2,k}, \quad -i\sqrt{\omega_k - \frac{k}{R}} u_{1,k}^* = \sqrt{\omega_k + \frac{k}{R}} u_{2,k}^*. \quad (38)$$

The most generic solution is

$$\mathbf{v}_k = e^{i\alpha_k} \begin{pmatrix} \sqrt{\omega_k + \frac{k}{R}} \\ i\sqrt{\omega_k - \frac{k}{R}} \end{pmatrix} \quad (39)$$

$$\mathbf{u}_k = e^{i\beta_k} \mathbf{v}_k. \quad (40)$$

Note that the feature to write $\mathbf{u} \propto \mathbf{v}^*$ is due to the specific choice of the Dirac representation. α and β are free parameter. One can set $\beta = \alpha = 0$, such that $\mathbf{u}_k = \mathbf{v}_k$.

4.2 Anticommutation Relations

With the momentum (respecting the Grassmann property[11])

$$\pi_a = -\frac{\partial \mathcal{L}}{\partial(\partial_\tau \psi_a)} = i\psi_a^\dagger, \quad (41)$$

the equal time anti-commutation relation for fermions (i.e. $\{\psi_a, \pi_b\} = i\delta_{ab}\delta(\theta_a - \theta_b)$) reads

$$\{\psi_a(\tau, \theta_x), \psi_b^\dagger(\tau, \theta_y)\} = \delta_{ab}\delta(\theta_x - \theta_y) \quad (42)$$

In order to project out the operators $c^{(\dagger)}$ and $d^{(\dagger)}$ from the fields, the properties

$$\begin{aligned} \mathbf{v}_k^\dagger \mathbf{v}_k &= \mathbf{v}_k^T \mathbf{v}_k^* = 2\omega_k \\ \mathbf{v}_k^\dagger \mathbf{v}_{-k}^* &= \mathbf{v}_k^T \mathbf{v}_{-k} = 0 \end{aligned} \quad (43)$$

are used.

The operators in Fourier space are

$$c_k = \int \frac{d\theta}{\sqrt{2\pi R} 2\omega_k N_k} \mathbf{v}_k^\dagger \psi e^{i(\omega_k \tau - k\theta/R)}, \quad d_k^\dagger = \int \frac{d\theta}{\sqrt{2\pi R} 2\omega_k N_k} \mathbf{v}_k^T \psi e^{-i(\omega_k \tau - k\theta/R)} \quad (44)$$

$$\text{and complex conjugated equations.} \quad (45)$$

Using the commutation relation eq.(42) gives then

$$\begin{aligned} \{c_k, c_l^\dagger\} &= \int \frac{d\theta_x d\theta_y}{2\pi R N_k N_l} \frac{(\mathbf{v}_k^\dagger)_a (\mathbf{v}_l^T)_b}{4\omega_k \omega_l} \{\psi_a^\dagger, \psi_b\} e^{i(\omega_k - \omega_l)\tau} e^{-i(k\theta_x - l\theta_y)/R} = \frac{\delta_{kl}}{2\omega_k N_k^2} \\ \Rightarrow \{c_k, c_l^\dagger\} &= \{d_k, d_l^\dagger\} = \delta_{kl} \\ \Rightarrow \{c_k, c_l\} &= \{c_k^\dagger, c_l^\dagger\} = \{d_k, d_l\} = \{d_k^\dagger, d_l^\dagger\} = \{c_k^{(\dagger)}, d_l^{(\dagger)}\} = 0, \quad \text{with } N_k = \frac{1}{\sqrt{2\omega_k}} \end{aligned} \quad (46)$$

Analogous to the boson field, The anticommutation relations are used to argue, that $c_k^\dagger$ and $d_k^\dagger$ acting as creation operators of particles and anti-particles respectively. c_k and d_k are annihilation operators. And N_k is chosen, such that the states are normalized to one.

The difference is, that interchanging two identical Grassmann variables creates a minus sign. Thus

$$a_k^\dagger a_k^\dagger |0\rangle = -a_k^\dagger a_k^\dagger |0\rangle \Rightarrow a_k^\dagger a_k^\dagger |0\rangle = 0. \quad (47)$$

and acting with a creation operator on a multi particle state, which already contains that particle gives zero (Pauli-exclusion principle).

4.3 Energy

The calculation of the free Hamiltonian H_0 is simplified, because inserting the e.o.m. eq.(33) in the Lagrangian eq.(17) gives

$$\mathcal{L}|_{EoM} = i\psi_1^\dagger \partial_+ \psi_1 + i\psi_2^\dagger \partial_- \psi_2 + im(\psi_1^\dagger \psi_2 - \psi_2^\dagger \psi_1) = i\psi_1^\dagger (-m\psi_2) + i\psi_2^\dagger (m\psi_1) + im(\psi_1^\dagger \psi_2 - \psi_2^\dagger \psi_1) = 0.$$

The Hamiltonian is

$$\begin{aligned} :H_0: &= \int d\theta : \pi \partial_\tau \psi : - \mathcal{L}|_{EoM} = i \int d\theta : \psi^\dagger \partial_\tau \psi : \\ &= \sum_{nm} \int \frac{-d\theta \omega_l}{2\pi R \sqrt{4\omega_k \omega_l}} : [\mathbf{v}_k^\dagger c_n^\dagger e^{i(\omega_k \tau - \frac{k}{R}\theta)} + \mathbf{v}_k^T d_k e^{-i(\omega_k \tau - \frac{k}{R}\theta)}] [-\mathbf{v}_l c_l e^{-i(\omega_l \tau - \frac{l}{R}\theta)} + \mathbf{v}_l^* d_l^\dagger e^{i(\omega_l \tau - \frac{l}{R}\theta)}] : \\ \Leftrightarrow :H_0: &= \sum_n \left[\frac{\mathbf{v}_k^\dagger \mathbf{v}_k}{2\omega_k} \omega_k c_k^\dagger c_k - \frac{\mathbf{v}_k^T \mathbf{v}_k^*}{2\omega_k} d_k d_k^\dagger + \mathbf{v}_k^\dagger \mathbf{v}_{-k}^* c_k^\dagger d_{-k} \dots + \mathbf{v}_k^T \mathbf{v}_{-k} d_k c_{-k}^\dagger \dots \right] \\ \Leftrightarrow :H_0: &= \sum_k \omega_k (c_k^\dagger c_k + d_k^\dagger d_k). \end{aligned} \quad (48)$$

In conclusion the free Spinor field can be written as:

$$\begin{aligned} \psi &= \sum_k \frac{1}{\sqrt{2\pi R} 2\omega_k} [\mathbf{v}_k c_k e^{-i(\omega_k \tau - k\theta/R)} + \mathbf{v}_k^* d_k^\dagger e^{i(\omega_k \tau - k\theta/R)}], \\ \text{with } \mathbf{v}_k &= \begin{pmatrix} \sqrt{\omega_k + \frac{k}{R}} \\ i\sqrt{\omega_k - \frac{k}{R}} \end{pmatrix}, \quad \omega_k = \sqrt{m^2 + \left(\frac{k}{R}\right)^2}. \end{aligned} \quad (49)$$

5 Interaction

5.1 Momentum and Energy Conservation

The solutions for the free fields always contain an exponential. In this section it will be shown that these exponentials do not appear in the Hamiltonian and can be treated separately.

It is used, that the free fields are written in a consistent way, such that creation- or annihilation operators are always multiplied by the same exponential:

$$\phi, \psi \propto a_k^\pm e^{\pm i(\omega_k \tau - k\theta/R)}.$$

Here, a^+/a^- is shorthand for any creation/annihilation operators of the fields.

The matrix element includes an integration over the angle, which results in the momentum conserving delta function. All interaction terms will be proportional to

$$H \propto \int d\theta a_n^\pm c_k^\pm d_l^\pm e^{i(\mp n \mp k \mp l)\theta/R} \quad (50)$$

$$= 2\pi R a_n^\pm c_k^\pm d_l^\pm \delta(\mp n \mp k \mp l) \quad (51)$$

The signs are linked to the creation/annihilation operators, e.g. one part is proportional to

$$H \propto 2\pi R a_n c_k^\dagger d_l^\dagger \delta(n - k - l)$$

The energy part of the exponentials cancel out in the end, because of Energy conservation³. In general

$$H \propto e^{i(\omega^{in} - \omega^{out})\tau} = 1. \quad (52)$$

So, the integral with all exponentials is replaced by $2\pi R$ times the delta function associated with momentum conservation.

5.2 Interaction Terms

The interaction Hamiltonian is given by $H = -\int d\theta \mathcal{L}_{int} = g \int d\theta \phi \bar{\psi} \psi$, as in eq.(15). A straight forward calculation gives

$$H = \sum_{nkl} \frac{g}{\sqrt{2\pi R} \cdot \Omega_n} (a_n^\dagger + a_{-n}) \cdot \begin{bmatrix} \delta(n - k + l) c_l^\dagger c_k \cdot \mathbf{v}_l^\dagger \gamma^0 \mathbf{v}_k / \sqrt{4\omega_k \omega_l} \\ \delta(n - k + l) d_k d_l^\dagger \cdot \mathbf{v}_k^T \gamma^0 \mathbf{v}_l^* / \sqrt{4\omega_k \omega_l} \\ + \delta(n - k + l) c_{-k}^\dagger d_l^\dagger \cdot \mathbf{v}_{-k}^\dagger \gamma^0 \mathbf{v}_l^* / \sqrt{4\omega_k \omega_l} \\ + \delta(n - k + l) d_k c_{-l} \cdot \mathbf{v}_k^T \gamma^0 \mathbf{v}_{-l} / \sqrt{4\omega_k \omega_l} \end{bmatrix} \quad (53)$$

With the definition eq.(49) in the appendix 18, it is shown that

$$\frac{\mathbf{v}_l^\dagger \gamma^0 \mathbf{v}_k}{\sqrt{4\omega_k \omega_l}} = -\frac{\mathbf{v}_k^T \gamma^0 \mathbf{v}_l^*}{\sqrt{4\omega_k \omega_l}} = \sqrt{\frac{2\omega_k \omega_l + 2m^2 - 2kl/R^2}{4\omega_k \omega_l}} \equiv \sqrt{\frac{1}{2} + M_{kl}} \quad (54)$$

$$\frac{\mathbf{v}_{-k}^\dagger \gamma^0 \mathbf{v}_l^*}{\sqrt{4\omega_k \omega_l}} = \frac{\mathbf{v}_k^T \gamma^0 \mathbf{v}_{-l}}{\sqrt{4\omega_k \omega_l}} = \text{sgn}(k + l) \sqrt{\frac{2\omega_k \omega_l - 2m^2 + 2kl/R^2}{4\omega_k \omega_l}} \equiv \text{sgn}(k + l) \sqrt{\frac{1}{2} - M_{kl}}, \quad (55)$$

³Energy is not conserved for single vertices in the particle basis chosen, but after diagonalization, energy is trivially conserved in the diagonalized basis (the in-states equal the out-states).

where

$$M_{kl} \equiv \frac{m^2 - kl/R^2}{2\omega_k\omega_l} \quad (56)$$

is used to build up all interactions. The term will make it easier to find symmetries, since

$$M_{k,l} = M_{-k,-l} = M_{l,k} \quad (57)$$

$$M_{kl} \in]-\frac{1}{2}, \frac{1}{2}] \quad (58)$$

$$M_{k,-k} = \frac{1}{2}, \quad \text{and} \quad M_{k,l} \rightarrow -\frac{1}{2}, \quad \text{for } k, l \gg m, \text{ or } k, l \ll m \quad (59)$$

Finally,

$$H = \sum_{k,l} \frac{g}{\sqrt{2\pi R \cdot 2\Omega_{k-l}}} (a_{k-l}^\dagger + a_{-(k-l)}) \left[\sqrt{\frac{1}{2} + M_{kl}} (c_l^\dagger c_k + d_l^\dagger d_k) \right. \\ \left. + \text{sgn}(k+l) \sqrt{\frac{1}{2} - M_{kl}} (c_{-k}^\dagger d_l^\dagger + d_k c_{-l}) \right], \quad (60)$$

and $M_{kl} = \frac{m^2 - kl/R^2}{2\omega_k\omega_l}$.

6 Conservation Laws

In this section, conserved quantities will be derived. The commutation relations will be shown to vanish, which justify[14] the use of eigenvectors of these operator as the Fock basis.

The free Hamiltonian did not impact the conclusions of the following chapters. For simplicity the notation $H_{int} \rightarrow H$ is used and the free Hamiltonian will be ignored here.

6.1 Momentum Conservation

As already mentioned in section 5.1, the Hamiltonian respects momentum conservation. The momentum operator is explicitly written as

$$\hat{L} = \sum_k k (a_k^\dagger a_k + c_k^\dagger c_k + d_k^\dagger d_k). \quad (61)$$

The basis in the Fock space will be constructed by creation/annihilation operators acting on the vacuum. Thereby the states are eigenvectors of $\hat{L}$. This choice is justified, since the Hamiltonian doesn't change the total momentum. Explicitly, one can show, that

$$[\hat{L}, H] = 0. \quad (62)$$

To derive the commutation relations, it is handy to use

$$\begin{aligned} [\hat{a}_k^\dagger \hat{a}_k, \hat{a}_n^\dagger] &= \delta_{kn} \hat{a}_k^\dagger \\ [\hat{a}_k^\dagger \hat{a}_k, \hat{a}_n] &= -\delta_{kn} \hat{a}_k^\dagger \\ [\hat{a}_k^\dagger \hat{a}_k, \hat{b}_n^\dagger] &= [\hat{a}_k^\dagger \hat{a}_k, \hat{b}_n] = 0, \end{aligned} \quad (63)$$

where $\hat{a}^{(\dagger)}$ can be the annihilation (creation) operator of any particle type and $\hat{b}^{(\dagger)}$ of any other kind. With that, it is straight forward to prove, that eq.(62) is indeed satisfied⁴.

6.2 Fermion Charge

This symmetry states that the number of fermions minus anti-fermions is conserved. The explicit form is

$$\hat{Q} = \sum_k (c_k^\dagger c_k - d_k^\dagger d_k). \quad (64)$$

Then, one can derive that (omitting the sums),

$$\left. \begin{aligned} [(c_k^\dagger c_k - d_k^\dagger d_k), c_n^\dagger c_m] &= (\delta_{kn} c_k^\dagger) c_m + c_n^\dagger (-\delta_{km} c_k) = 0 \\ [(c_k^\dagger c_k - d_k^\dagger d_k), d_n^\dagger c_m] &= -(-\delta_{kn} d_k) c_m + d_n^\dagger (-\delta_{km} c_m) = 0 \\ [(c_k^\dagger c_k - d_k^\dagger d_k), c_n^\dagger d_m^\dagger] &= (\delta_{kn} c_k^\dagger d_m^\dagger - c_n^\dagger (\delta_{km} d_m^\dagger)) = 0 \end{aligned} \right\} \Rightarrow [\hat{Q}, H] = 0 \quad (65)$$

$$[(c_k^\dagger c_k - d_k^\dagger d_k), c_n^\dagger c_n] = (\delta_{kn} c_k^\dagger) c_n + c_n^\dagger (-\delta_{kn} c_k) = 0 \Rightarrow [\hat{Q}, \hat{L}] = 0. \quad (66)$$

7 Particle Basis

The Fock basis used will be described here. It is then straight forward to outline the implementation.

A way to create the basis, is to define the states as any number and combinations of creation operators, acting on the vacuum in the form of

$$|i\rangle = \{a_k^\dagger, c_n^\dagger, d_m^\dagger\} |0\rangle, \quad (67)$$

where the bracket $\{ \cdot \}$ indicates all combinations of creation operators.

$|i\rangle$ is an arbitrary short name for a particular basis state. Here, it refers to a generic basis state, but for calculations one has to apply an explicit definition, e.g. one could define $|0\rangle \equiv |0\rangle$, $|1\rangle \equiv a_0^\dagger |0\rangle$ and so on. A more detailed example is written out in chapter 9.1.

Nevertheless, there are some peculiarities, how the operators act on the vacuum.

Bosonic Part:

Unitary normalization is used, which imply additional normalization factors, when boson operators act on states (see eq.(28)).

Thus, a state $|i\rangle$ contains also a normalization factor beside the creation operators. However, the action of a_k or $a_k^\dagger$ as they appear in the Hamiltonian is completely defined in eq.(28). In the derivation, the exact normalization factor is not needed. Henceforth, the factor will be omitted and all states are assumed to automatically satisfying the normalization eq.(27).

Fermionic Part:

⁴One example:

$$\begin{aligned} [\hat{L}, a_{n-m}^\dagger c_{-n}^\dagger d_m^\dagger] &= [\hat{L}, a_{n-m}^\dagger] c_{-n}^\dagger d_m^\dagger + a_{n-m}^\dagger [\hat{L}, c_{-n}^\dagger] d_m^\dagger + a_{n-m}^\dagger c_{-n}^\dagger [\hat{L}, d_m^\dagger] \\ &= \sum_k \left([k a_k^\dagger a_k, a_{n-m}^\dagger] c_{-n}^\dagger d_m^\dagger + a_{n-m}^\dagger [k c_k^\dagger c_k, c_{-n}^\dagger] d_m^\dagger + a_{n-m}^\dagger c_{-n}^\dagger [k d_k^\dagger d_k, d_m^\dagger] \right) \\ &= ((n-m) + (-n) + m) a_{n-m}^\dagger c_{-n}^\dagger d_m^\dagger = 0. \end{aligned}$$

The other parts are calculated in a similar way.

The anticommutation relations lead to the Pauli exclusion principle as described in eq.(47). Moreover, the order in which fermion operators act on the vacuum needs to be defined. Take for example a multiparticle state $|i\rangle$, which contains one Fermion c and one anti-fermion d . Then, if one define

$$c^\dagger d^\dagger |0\rangle \equiv |i\rangle, \text{ then } d^\dagger c^\dagger |0\rangle = -|i\rangle.$$

So, it has to be defined, which order gives a minus sign.

7.1 Ordering and Occupation Number Representation

Multiple creation operators, which act on the vacuum, are associated to a basis state (ket-vector), if they are ordered in the following way:

1. The ordering of particle types is (from left to right): bosons ($a^\dagger$); fermions ($c^\dagger$); anti-fermions ($d^\dagger$).
2. Absolute momentum $|n|$ of creation operators is ascending from left to right. (68)
3. ($n > 0$)-states are on the left of ($n < 0$)-states.

Where the priority lies in the first point and then descending. Since the norm of a state needs to be positive, In the definition of bra-vectors, annihilation operators have to be ordered in the opposite way of creation operators.

To be consistent also inside other operators, the creation operators are always ordered in this way and annihilation operators in the opposite way, as for example in H in eq.(60).

The occupation number representation (ONR)[12] can be effortlessly defined in the same ordering. The number of particles written in the slots are associated with the momentum

$$\{0_a, 1_a, -1_a, 2_a, \dots ; 0_c, 1_c, -1_c, 2_c, \dots ; 0_d, 1_d, -1_d, 2_d, \dots\},$$

where the index indicates the particle type. Or, to be precise, the multiplicity of the operator $a_k^\dagger$ shall be named Ak and similar Ck and Dk for (anti-)fermions (such that e.g. $Ck \in \{0, 1\}$). Then a basis state is generally written as

$$|i\rangle \equiv [a_0^\dagger]^{A0} [a_1^\dagger]^{A1} \dots [c_0^\dagger]^{C0} [c_1^\dagger]^{C1} \dots [d_0^\dagger]^{D0} [d_1^\dagger]^{D1} \dots |0\rangle, \quad (69)$$

which in the ONR is written as

$$|i\rangle \hat{=} \{A0, A1, \dots ; C0, C1, \dots ; D0, D1, \dots\}. \quad (70)$$

7.2 Momentum and Fermion Charge Conservation

The basis states are eigenvectors to the momentum and the fermion charge operator. Since the Hamiltonian respects these symmetries, a state with certain momentum eigenvalue and fermion charge eigenvalue can only interact with states, which have the same quantum numbers. Explicitly

$$\langle i|H|j\rangle = 0, \text{ if } \langle i|\hat{L}|i\rangle \neq \langle j|\hat{L}|j\rangle, \text{ or } \langle i|\hat{Q}|i\rangle \neq \langle j|\hat{Q}|j\rangle.$$

With that, H_{ij} can be easily rewritten in a block diagonal, where each block is associated with a certain set of quantum numbers $\hat{L}$ and $\hat{Q}$. Each block can be diagonalized separately then.

In focus is the submatrix corresponding to total momentum and fermion charge of zero, because this block contains the vacuum $|0\rangle$.

The basis is restricted by these eigenvalues, thus

$$|i\rangle \in \{|j\rangle \mid \langle j|\hat{L}|j\rangle = 0 \wedge \langle j|\hat{Q}|j\rangle = 0\}. \quad (71)$$

In appendix 17 is a detailed description, how such a basis could be generated.

8 Symmetries of the Hamiltonian

In principle, it is possible to start implementing H , as in eq.(60). But there are additional symmetries, which reduce the number of elements to calculate.

8.1 Hermitian Symmetry

It is straight forward to show⁵, that the Hamiltonian is Hermitian. In particular, one can decompose

$$H = h + h^\dagger, \text{ with} \\ h^\dagger = \sum_{n,m} \frac{ga_{n-m}^\dagger}{2\sqrt{\pi R\Omega_{n-m}}} \left[\sqrt{\frac{1}{2} + M_{nm}} (c_m^\dagger c_n + d_m^\dagger d_n) + \text{sgn}(n+m) \sqrt{\frac{1}{2} - M_{nm}} (c_{-n}^\dagger d_m^\dagger + d_n c_{-m}) \right]. \quad (72)$$

This will half the calculation, since only $h^\dagger$ is needed in order to calculate all Matrix elements, since

$$\begin{aligned} h_{ij}^\dagger &\equiv \langle i|h^\dagger|j\rangle \quad \Leftrightarrow \quad h_{ij} = h_{ji}^\dagger \\ H_{ij} &= h_{ij}^\dagger + h_{ji}^\dagger. \end{aligned} \quad (73)$$

Thus $h^\dagger$ is used to create half the elements, while the other half is generated by switching the incoming and outgoing states.

8.2 Charge Conjugation

The charge conjugation operator transforms all particles into their anti-particles. The operator is defined through the following properties:

$$\begin{aligned} \hat{C}a_n^\dagger\hat{C} &= a_n^\dagger, & \hat{C}a_n\hat{C} &= a_n \\ \hat{C}c_n^\dagger\hat{C} &= d_n^\dagger, & \hat{C}c_n\hat{C} &= d_n \\ \hat{C}d_n^\dagger\hat{C} &= c_n^\dagger, & \hat{C}d_n\hat{C} &= c_n \\ \hat{C}^\dagger &= \hat{C}, \quad \hat{C}\hat{C} = 1, \quad \hat{C}|0\rangle = |0\rangle. \end{aligned} \quad (74)$$

The commutation relation with other conserved quantities are

$$\hat{C} (a_n^\dagger a_n + c_n^\dagger c_n + d_n^\dagger d_n) \hat{C} = (a_n^\dagger a_n + c_n^\dagger c_n + d_n^\dagger d_n) \Rightarrow [\hat{C}, \hat{L}] = 0 \quad (75)$$

$$\hat{C} (c_n^\dagger c_n - d_n^\dagger d_n) \hat{C} = -(c_n^\dagger c_n - d_n^\dagger d_n) \Rightarrow \{\hat{C}, \hat{Q}\} = 0. \quad (76)$$

The fermion charge switches sign under charge conjugation. But, states can be eigenvectors of both operators, if the eigenvalue of $\hat{Q}$ is zero. So, it can work for the chosen basis.

⁵The part which is missing in $h^\dagger$ w.r.t. H in eq.(60), is indeed h . For the derivation $n \leftrightarrow m$ needs to be renamed by using $M_{nm} = M_{mn}$.

To show, that the interaction respects charge conjugation the Hamiltonian $H = h^\dagger + h$ will be redefined as

$$\begin{aligned} h^\dagger &= \mathcal{H} + \mathcal{H}^c + \mathcal{H}^0, \\ \text{with } \hat{C}\mathcal{H}^0\hat{C} &\equiv \mathcal{H}^0 \text{ and } \hat{C}\mathcal{H}\hat{C} \equiv \mathcal{H}^c. \end{aligned} \quad (77)$$

As shown in eq.(74), $\hat{C}$ acts on an operator in the same way as on the complex conjugate. Assuming eq.(77), it follows that

$$[\hat{C}, H] = 0 \quad (78)$$

In the following the split eq.(77) will be shown to be valid.

T-Channel:

The "T-channel" refers to the part $\propto a_{n-m}^\dagger (c_m^\dagger c_n + d_m^\dagger d_n)$. The naming is due to the fact, that the $2 \rightarrow 2$ T-channel Feynman diagram would be build up by two times this Vertex. Since this part trivially respects charge conjugation, it can be separated as

$$\begin{aligned} h^\dagger &\subset \sum_{n,m} \frac{ga_{n-m}^\dagger}{2\sqrt{\pi R\Omega_{n-m}}} \sqrt{\frac{1}{2} + M_{nm}} (c_m^\dagger c_n + d_m^\dagger d_n) \\ \Rightarrow \begin{cases} \mathcal{H} &\subset \sum_{n \neq m} \frac{ga_{n-m}^\dagger}{2\sqrt{\pi R\Omega_{n-m}}} \sqrt{\frac{1}{2} + M_{nm}} c_m^\dagger c_n \\ \mathcal{H}^0 &= \frac{ga_0^\dagger}{2\sqrt{\pi R\Omega_0}} \hat{M}, \end{cases} \end{aligned} \quad (79)$$

$$\text{with } \hat{M} \equiv \sum_n \sqrt{\frac{1}{2} + M_{nn}} (c_n^\dagger c_n + d_n^\dagger d_n) \quad (80)$$

The operator $\hat{M}$ acts as a weighted counting of fermions. It is the only vertex which leaves the fermion part unchanged. To derive the normalization this part needs to be treated separately.

S-Channel and Z-Channel:

The "S-channel" refers to the vertex $\propto a_{n-m}^\dagger d_n c_{-m}$ and the "Z-channel"⁶ to $\propto a_{n-m}^\dagger c_{-n}^\dagger d_m^\dagger$. In order to split these into two parts, the sum is halved. Instead of summing over all $\{n, m\}$, it is split into $\{-n < m\}$ and $\{-n > m\}$ (while $-n = m$ gives zero for these channels). In appendix 18 it is shown, that these parts are indeed charge conjugates of each other. So,

$$\begin{aligned} h^\dagger &\subset \sum_{n,m} \frac{ga_{n-m}^\dagger}{2\sqrt{\pi R\Omega_{n-m}}} \text{sgn}(n+m) \sqrt{\frac{1}{2} - M_{nm}} (c_{-n}^\dagger d_m^\dagger + d_n c_{-n}) \\ \Rightarrow \mathcal{H} &\subset \sum_{-n < m} \frac{ga_{n-m}^\dagger}{2\sqrt{\pi R\Omega_{n-m}}} \text{sgn}(n+m) \sqrt{\frac{1}{2} - M_{nm}} (c_{-n}^\dagger d_m^\dagger + d_n c_{-n}) \end{aligned} \quad (81)$$

⁶The $2 \rightarrow 2$ S-channel in time ordered perturbation theory corresponds to two diagrams in old fashioned perturbation theory[15]. The one, where the intermediate particle travels forward in time, still looks like the S-channel. The other diagram, has a "Z" shape, as the intermediate particle travels backwards. The type of vertices which build up the S-channel-shaped diagram are called "S-channel" here. And the vertices of the Z-shaped diagram "Z-channel".

In summery, for the next chapters the following definitions are used

$$H_{int} = h^\dagger + h, \text{ with } h^\dagger = \mathcal{H}^0 + \mathcal{H} + \mathcal{H}^c$$

$$\mathcal{H}^0 = \frac{ga_0^\dagger}{2\sqrt{\pi R\Omega_0}}\hat{M}, \text{ with } \hat{M} = \sum_n \sqrt{\frac{1}{2} + M_{nn}} (c_n^\dagger c_n + d_n^\dagger d_n) \quad (82)$$

$$\mathcal{H} = \frac{g}{2\sqrt{\pi R}} \left[\sum_{n \neq m} \frac{a_{n-m}^\dagger}{\sqrt{\Omega_{n-m}}} \sqrt{\frac{1}{2} + M_{nm}} c_m^\dagger c_n + \sum_{-n < m} \frac{\text{sgn}(n+m)a_{n-m}^\dagger}{\sqrt{\Omega_{n-m}}} \sqrt{\frac{1}{2} - M_{nm}} (c_{-n}^\dagger d_m^\dagger + d_n c_{-n}) \right] \quad (83)$$

$$\mathcal{H}^c = \frac{g}{2\sqrt{\pi R}} \left[\sum_{n \neq m} \frac{a_{n-m}^\dagger}{\sqrt{\Omega_{n-m}}} \sqrt{\frac{1}{2} + M_{nm}} d_m^\dagger d_n + \sum_{-n < m} \frac{\text{sgn}(n+m)a_{n-m}^\dagger}{\sqrt{\Omega_{n-m}}} \sqrt{\frac{1}{2} - M_{nm}} (d_{-n}^\dagger c_m^\dagger + c_n d_{-n}) \right] \quad (84)$$

Therefore, charge conjugation can be used to build up the basis. Note that the states described in chapter 7 are not symmetric under charge conjugation, but the eigenvectors of $\hat{C}$ are build up, using the described "full basis states". In doing so, the implementation is described without charge conjugation first.

9 Implementation

In this section it will be illustrated, how one could calculate the matrix elements $H_{ij} = \langle i|H|j \rangle$. The idea is to clarify the first steps to make the implementation of charge conjugation easier to understand. The coding is done using Mathematica.

9.1 Basis

The Fock basis and the occupation number representation (ONR) is introduced in section 7. The way to implement this basis is to generate all possible ONR with total momentum and fermion charge equal zero. Once the basis is complete, each state gets a position number, which defines the matrix representation H_{ij} of the Hamiltonian. Additional properties like the free energy, could also be gathered in each state. For example the form

$$\{\text{ONR, position number, energy}\}$$

is used here. Note that, if the position numbers are ordered according to the free energy, a sub-matrix of a higher truncation matrix gives rise to a lower truncation matrix. The energy of the state, associated with the right most column of the Hamiltonian is the truncation energy.

For example for $M = 1$, $m = 0.3$ and $R = 1$, the first basis states would look like

$$\begin{aligned} \{\{0, \dots; 0, \dots; 0, \dots\}, 1, 0\} &\leftrightarrow |0\rangle \\ \{\{0, \dots; 1, 0, \dots; 1, 0, \dots\}, 2, 0.6\} &\leftrightarrow c_0^\dagger d_0^\dagger |0\rangle \\ \{\{1, 0, \dots; 0, \dots; 0, \dots\}, 3, 1\} &\leftrightarrow a_0^\dagger |0\rangle \\ \{\{2, 0, \dots; 0, \dots; 0, \dots\}, 4, 2\} &\leftrightarrow \frac{1}{\sqrt{2}} a_0^\dagger a_0^\dagger |0\rangle \\ \{\{0, \dots; 0, 1, 0, \dots; 0, 0, 1, 0, \dots\}, 5, 2.09\} &\leftrightarrow c_1^\dagger d_{-1}^\dagger |0\rangle \\ \{\{0, \dots; 0, 0, 1, 0, \dots; 0, 1, 0, \dots\}, 6, 2.09\} &\leftrightarrow c_{-1}^\dagger d_1^\dagger |0\rangle. \end{aligned} \quad (85)$$

9.2 Matrix Elements

The matrix elements are calculated by working off each channel and each combination of momenta one by one, yielding a nested list of elements in the form

$$\{i, j, H_{ij}\},$$

which defines the Hamiltonian. As an example, the shown basis states – i.e. a truncation of $E_{max} = 2.09$ – and the S-channel-Vertex (see chapter 8.2) is used. The operators in this vertex are $a_{n-m}^\dagger d_n c_{-m}$, thus the cases $a_0^\dagger d_{-1} c_1$ and $a_0^\dagger d_1 c_{-1}$ are non-vanishing. In particular $a_0^\dagger d_{-1} c_1$ shall be used. The calculation of each vertex is done in three steps.

First, the in-states have to be found. This are the basis states, which are still below the truncation, after the considered part of the Hamiltonian acts on it. So, in this case

$$E + \Omega_0 - \omega_1 - \omega_{-1} \leq 2.09.$$

Additionally the states must not give zero after the creation/annihilation operators act on it. That means selecting those states, which contain the incoming particles and none of the outgoing fermions (Pauli exclusion principle, see eq.(47)).

In this case, the slot of the ONR associated with $c_1^\dagger$ and $d_{-1}^\dagger$ must both equal 1.

Their position numbers form the first entries of each element. There is only one state satisfying the conditions. The matrix element is then

$$\{5, \cdot, \cdot\},$$

which states the initial state.

The second number is given by the out-states. They are calculated by modifying each ONR, according to the Hamiltonian element and finding the position number associated to the new ONR. In doing so, the Hash Mechanism[7] can be used, e.g. using an Association in Mathematica. The ONR can be used as a key, which returns the new position number directly. This mechanism didn't scale with the number of elements⁷, which makes it advantageous for large data sets.

In the example, slot $a_0^\dagger$ in the ONR of state 5 is raised and slot $c_1^\dagger$ and $d_{-1}^\dagger$ are lowered, leaving the ONR $\{1, 0, \dots\}$. The returned position number is 3. So, the matrix element is

$$\{5, 3, \cdot\}.$$

Lastly, the amplitude is calculated according to the occupations of the in-states. One has to keep in mind that the bosonic operators give an additional normalization as described in eq.(28). And fermions may give a minus sign due to the anticommutation relation:

Suppose a fermion operator acts on slot i in the ONR. The sign is then given by $(-1)^n$, where n is the number of fermions with a slot number $j < i$. If multiple, proper ordered fermions act on a state. The acting operators itself are not counted.

In the ONR of state 5, before the $c_1^\dagger$ -slot is no fermion inscribed, and before the $d_{-1}^\dagger$ -slot is no other fermion than $c_1^\dagger$. The normalization is +1. The entries are therefore

$$\{5, 3, \frac{g \operatorname{sgn}(-1-1)}{2\sqrt{\pi R \Omega_0}} \sqrt{\frac{1}{2} - M_{-1,-1}}\}.$$

⁷Another way is to search directly in the basis for the new ONR. In Mathematica it would be done by the command "Position". This command is inefficient for large data sets, as the timing scales linear with the number of basis states (or logarithmically using binary search). For the thesis, it was programmed both ways. Searching the basis for the ONRs had a runtime around 20h for certain parameter, while a code using hashing needs 8min, giving the same result.

It is straight forward to transform the elements in a symmetric Hamiltonian matrix. The total Hamiltonian of $M = 1$, $m = 0.3$, $R = 1$, $E_{max} = 2.09$ is

$$H_{ji} = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0.6 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & -0.27 & 0.27 \\ 0 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & -0.27 & 0 & 2.09 & 0 \\ 0 & 0 & 0.27 & 0 & 0 & 2.09 \end{pmatrix}. \quad (86)$$

9.3 Charge Conjugation

As shown in chapter 8.2, even and odd states don't interact. Thus, the size of the Hamiltonian can be halved. The vacuum has a +1 eigenvalue, so the focus lies on the C-even states.

The difficulty is, that many eigenvectors of $\hat{C}$ contain two of the full-basis states. The new basis will be constructed from the old, using the notation

$$|\alpha\rangle = \begin{cases} |i\rangle, & \text{if } \hat{C}|i\rangle = |i\rangle \\ \frac{1}{\sqrt{2}}(|i\rangle + \hat{C}|i\rangle) \equiv \frac{1}{\sqrt{2}}(|i\rangle + |\bar{i}\rangle) & \text{if } \hat{C}|i\rangle \neq |i\rangle, \end{cases} \quad (87)$$

where a Greek letter indicates a generic state of the c-basis and Latin indices a full-basis state. The two cases are referred to as singlet and doublet respectively. Moreover, the first doublet state will be called reference state, which is chosen to appear with a plus sign in the basis. $|\bar{i}\rangle$ on the other hand - the secondary states - may get a minus sign, since the fermions must be reordered according to eq. (68). It is straight forward to show, that

$$|\bar{i}\rangle = (-1)^{\sum_k c_k^\dagger c_k} |j\rangle,$$

where $|j\rangle$ is a properly ordered state. It is used, that the number of fermions equal the number of anti-fermions in the chosen basis. The fermion counting operator could as well run over anti-fermions instead.

The difficulty is to implement doublets. One solution is to calculate the matrix elements of the c-basis by using full-basis states. For that the in-states are translated to full-basis states and then the out-states are translated back. So, $H_{\beta\alpha}$ is written in terms of H_{ji} , formally:

$$H_{\beta\alpha} = \sum_{i,j} M_j D_{\beta j} H_{ji} C_{i\alpha} N_\alpha, \quad (88)$$

where $C_{i\alpha}$ and $D_{\beta j}$ define the basis transitions, but normalization factors N_α and M_j are treated separately.

9.3.1 Charge Even Basis

$|\alpha\rangle = |i\rangle C_{i\alpha}$ is no more than a formal definition. It is equivalent to represent the doublets by the ONR of the reference state only. The information of the secondary state is defined by charge conjugation. The singlets are written as before.

The basis states still need to be extended by two entries in order to keep track of the normalization as presented in eq. (87). Multiplying this factor by $\sqrt{2}$ lead directly to the normalization factors N and M of eq. (88). This will be derived in the next section, but here it can be viewed as an equivalent way to store the information. Therefore, the reference state gets a 1 inscribed and the next entry of the secondary state get the eigenvalue of $(-1)^{\sum_k c_k^\dagger c_k}$.

For singlets the factor is $\sqrt{2}$, but the eigenvalues of the C-operator are +1 or -1 for these. So, some states are C-odd and not in the C-basis. They get a position number and normalization of zero. Using the same parameter as before, the first few states are then

$$\begin{aligned}
& \{ \{0, \dots; 0, \dots; 0, \dots\}, 1, \quad 0, \sqrt{2}, \sqrt{2} \} \leftrightarrow |0\rangle \\
& \{ \{0, \dots; 1, 0, \dots; 1, 0, \dots\}, 0, \quad 0.6, \quad 0, \quad 0 \} \leftrightarrow c_0^\dagger d_0^\dagger |0\rangle \\
& \{ \{1, 0, \dots; 0, \dots; 0, \dots\}, 2, \quad 1, \sqrt{2}, \sqrt{2} \} \leftrightarrow a_0^\dagger |0\rangle \\
& \{ \{2, 0, \dots; 0, \dots; 0, \dots\}, 3, \quad 2, \sqrt{2}, \sqrt{2} \} \leftrightarrow a_0^\dagger a_0^\dagger |0\rangle \\
& \{ \{0, \dots; 0, 1, 0, \dots; 0, 0, 1, 0, \dots\}, 4, 2.09, \quad 1, \quad -1 \} \leftrightarrow \frac{1}{\sqrt{2}}(c_1^\dagger d_{-1}^\dagger - c_{-1}^\dagger d_1^\dagger) |0\rangle
\end{aligned} \tag{89}$$

where the second normalization factor for singlets is to keep the shape consistent. Furthermore, state number 3 would get a normalization factor of $1/\sqrt{2}$ due to the two bosons. This factor is already taken into account and can be omitted in order to be not confused with the factors coming from doublets.

Thereby the transition $\alpha \rightarrow i \hat{=} C_{i\alpha}$ is defined, where α is written as a position number, and i as a ONR. Comparing to the formal expression eq.(88), the part $|i\rangle C_{i\alpha}$ on the r.h.s. is given by the first entries of the states.

Similar $D_{\beta j} \hat{=} j \rightarrow \beta$ or $\langle \beta | = D_{\beta j} \langle j |$ turns the ONR to a position number. By using an Association (hash mechanism) $M_j D_{j\beta}$ is written as

$$\text{ONR}(j) \rightarrow \{\text{pos. nr.}(j), M_j\}.$$

In contrast to the in-states, all full-basis states are needed, as $\langle j | H | i \rangle$ may return the ONR of the secondary states or charge odd singlets.

The first entries are given by the ONRs with the corresponding position numbers present in the C-basis eq.(89), also including the c-odd singlets. Moreover, the M_j entries are given by the first normalization factors. Additionally the second doublet states get entries, where the charge conjugated ONR is used as a key, which returns the same position number of the reference state, together with the second normalization factor (since this state might be defined with a minus in the chosen basis). For the example states this give

$$\begin{aligned}
& \{0, \dots; 0, \dots; 0, \dots\} \rightarrow \{1, \sqrt{2}\} \\
& \{0, \dots; 1, 0, \dots; 1, 0, \dots\} \rightarrow \{0, 0\} \\
& \{1, 0, \dots; 0, \dots; 0, \dots\} \rightarrow \{2, \sqrt{2}\} \\
& \{2, 0, \dots; 0, \dots; 0, \dots\} \rightarrow \{3, \sqrt{2}\} \\
& \{0, \dots; 0, 1, 0, \dots; 0, 0, 1, 0, \dots\} \rightarrow \{4, 1\} \\
& \{0, \dots; 0, 0, 1, 0, \dots; 0, 1, 0, \dots\} \rightarrow \{4, -1\}.
\end{aligned} \tag{90}$$

Of course, the C-odd singlets must not appear in $H_{\beta\alpha}$ and have to be sorted out later.

In the next chapters, it is explained, how to derive the elements $H_{\beta\alpha}$, using the old basis elements H_{ji} , which lead to the normalization factors N_α and M_j .

9.3.2 Matrix Elements of $\mathcal{H}^0$

For the derivation, only the general form of the states as in eq.(87) is needed. In there, the full-basis states are taken to be normalized, but additional specifications are not needed.

The matrix elements

$$\mathcal{H}_{\beta\alpha}^0 = g/2\sqrt{\pi R\Omega_0} \cdot \langle \beta | a_0^\dagger \hat{M} | \alpha \rangle, \quad \text{with } \hat{M} = \sum_n \sqrt{1/2 + M_{nn}} (c_n^\dagger c_n + d_n^\dagger d_n),$$

leave the fermionic part unchanged. Therefore, the transitions are singlet to singlet, or doublet to doublet, but no interchange.

First, let $|\alpha\rangle = |a\rangle$ and $|\beta\rangle = |b\rangle$ be two singlets, then

$$\mathcal{H}_{\beta\alpha}^0 = \langle b|\mathcal{H}^0|a\rangle. \quad (91)$$

These elements are generated exactly as before (without further normalization).

For two doublets $|\alpha\rangle = 1/\sqrt{2}(|a\rangle + |\bar{a}\rangle)$ and $|\beta\rangle = 1/\sqrt{2}(|b\rangle + |\bar{b}\rangle)$ it follows, that

$$\mathcal{H}_{\beta\alpha}^0 = \frac{1}{2} (\langle b|\mathcal{H}^0|a\rangle + \langle \bar{b}|\mathcal{H}^0|\bar{a}\rangle + \langle \bar{b}|\mathcal{H}^0|a\rangle + \langle b|\mathcal{H}^0|\bar{a}\rangle) = \langle b|\mathcal{H}^0|a\rangle + \langle \bar{b}|\mathcal{H}^0|a\rangle. \quad (92)$$

Charge conjugation symmetry is applied twice; e.g.

$$\langle \bar{b}|\mathcal{H}^0|\bar{a}\rangle = \langle \bar{b}|\hat{C}\hat{C}\mathcal{H}^0\hat{C}\hat{C}|\bar{a}\rangle = \langle b|\mathcal{H}^0|a\rangle.$$

Then, only C-basis states are taken as in-states, hence $\mathcal{H}^0$ acts on the whole C-basis - implemented as described before. But, by using charge conjugation the secondary states are not needed.

For the outgoing state on the other hand $\mathcal{H}^0|a\rangle$ could introduce a transition to either a reference state $|b\rangle$ or to a secondary state $|\bar{b}\rangle$.

For example $\mathcal{H}^0$ acting on state 4 in eq.(89) would give

$$\mathcal{H}^0 \frac{1}{\sqrt{2}}(c_1^\dagger d_{-1}^\dagger - c_{-1}^\dagger d_1^\dagger)|0\rangle \propto \frac{1}{\sqrt{2}}(a_0^\dagger c_1^\dagger d_{-1}^\dagger - a_0^\dagger c_{-1}^\dagger d_1^\dagger)|0\rangle = -\frac{1}{\sqrt{2}}(a_0^\dagger c_{-1}^\dagger d_1^\dagger - a_0^\dagger c_1^\dagger d_{-1}^\dagger)|0\rangle,$$

but it is not defined, whether the first or the second state is chosen to be the reference state. Considering just the reference in-state give

$$\mathcal{H}^0 c_1^\dagger d_{-1}^\dagger|0\rangle \propto a_0^\dagger c_1^\dagger d_{-1}^\dagger|0\rangle,$$

which could be a secondary state. In this case the state may appear with a minus sign in the chosen basis. Thus, the sign is given by the second column of normalization factors in eq.(89).

Depending on the basis choice, one term in eq.(92) is zero, so $\mathcal{H}_{\beta\alpha}^0 = \mathcal{H}_{ba}^0$, or $\mathcal{H}_{\beta\alpha}^0 = \mathcal{H}_{\bar{b}\bar{a}}^0$ (what ever is unequal zero). Hence the normalization is one. In conclusion

$$\mathcal{H}_{\beta\alpha}^0 = \sum_{j,a} \text{sgn}(M_j) D_{\beta j} \mathcal{H}_{ja}^0 C_{a\alpha} \quad (93)$$

where $|a\rangle C_{a\alpha}$ is given by the ONR of the C-basis states in eq.(89). $\langle\beta| = D_{\beta j}\langle j|$ and $\text{sgn}(M_j)$ are received by feeding the ONR of $\mathcal{H}^0|a\rangle$ in the Association eq.(90) and acting with $\text{sign}()$ on the second entry.

9.3.3 Matrix Elements of $\mathcal{H} + \mathcal{H}^c$

Taking $\mathcal{H}^0$ as an example, the other cases are treated exactly in the same way. The doublet-doublet transition of $\mathcal{H} + \mathcal{H}^c$ from $|\alpha\rangle = 1/\sqrt{2}(|a\rangle + |\bar{a}\rangle)$ to $|\beta\rangle = 1/\sqrt{2}(|b\rangle + |\bar{b}\rangle)$ give eight terms with normalization $1/2$. Analogous to the previous case the factor cancels, as the number of terms is halved to

$$\begin{aligned} [\mathcal{H} + \mathcal{H}^c]_{\beta\alpha} &= \frac{1}{2} [(\mathcal{H}_{ba} + \mathcal{H}_{\bar{b}\bar{a}}^c) + (\mathcal{H}_{\bar{b}a} + \mathcal{H}_{b\bar{a}}^c) + (\mathcal{H}_{ba}^c + \mathcal{H}_{\bar{b}\bar{a}}) + (\mathcal{H}_{\bar{b}a}^c + \mathcal{H}_{b\bar{a}})] \\ \Leftrightarrow [\mathcal{H} + \mathcal{H}^c]_{\beta\alpha} &= \mathcal{H}_{ba} + \mathcal{H}_{\bar{b}\bar{a}}^c + \mathcal{H}_{ba}^c + \mathcal{H}_{\bar{b}\bar{a}}, \quad \text{for } |a\rangle \neq |\bar{a}\rangle, \text{ and } |b\rangle \neq |\bar{b}\rangle. \end{aligned} \quad (94)$$

It is straight forward to show, that no more than one term can be non zero⁸. Hence, the normalization factor is one, just as for the $\mathcal{H}^0$ -doublet transition.

⁸Assuming $\mathcal{H}_{ba} \neq 0 \Rightarrow |a\rangle \propto \mathcal{H}|b\rangle$. But since neither of the states $\mathcal{H}|\bar{b}\rangle$, $\mathcal{H}^c|b\rangle$ and $\mathcal{H}^c|\bar{b}\rangle$ is proportional to $\mathcal{H}|b\rangle$, the other elements $\mathcal{H}_{\bar{b}a}^c$, $\mathcal{H}_{ba}^c$ and $\mathcal{H}_{\bar{b}\bar{a}}^c$ must be zero.

The doublet-singlet transition from $|\alpha\rangle = 1/\sqrt{2}(|a\rangle + |\bar{a}\rangle)$ to $|\beta\rangle = |b\rangle$ give four terms with normalization $1/\sqrt{2}$. Using the symmetry as before leave a factor of $\sqrt{2}$, so

$$\begin{aligned} [\mathcal{H} + \mathcal{H}^c]_{\beta\alpha} &= \frac{1}{\sqrt{2}} [(\mathcal{H}_{ba} + \mathcal{H}_{b\bar{a}}^c) + (\mathcal{H}_{ba}^c + \mathcal{H}_{\bar{b}a})] \\ \Leftrightarrow [\mathcal{H} + \mathcal{H}^c]_{\beta\alpha} &= \begin{cases} \sqrt{2}(\mathcal{H}_{ba} + \mathcal{H}_{ba}^c), & \text{for } |a\rangle \neq |\bar{a}\rangle, \text{ and } |b\rangle = |\bar{b}\rangle \\ 0, & \text{for } |a\rangle \neq |\bar{a}\rangle, \text{ and } |b\rangle = -|\bar{b}\rangle. \end{cases} \end{aligned} \quad (95)$$

In the second case, there are cancellations between the terms, since $\mathcal{H}_{ba} = -\mathcal{H}_{b\bar{a}}^c$. But because the implementation uses just the reference state $|a\rangle$ the cancellation is only visible through the appearance of a C-odd outgoing state.

In the singlet-doublet transition from $|\alpha\rangle = |a\rangle$ to $|\beta\rangle = 1/\sqrt{2}(|b\rangle + |\bar{b}\rangle)$, the symmetry is used slightly differently, as

$$\begin{aligned} [\mathcal{H} + \mathcal{H}^c]_{\beta\alpha} &= \frac{1}{\sqrt{2}} (\mathcal{H}_{ba} + \mathcal{H}_{b\bar{a}}^c + \mathcal{H}_{\bar{b}a} + \mathcal{H}_{ba}^c) \\ \Leftrightarrow [\mathcal{H} + \mathcal{H}^c]_{\beta\alpha} &= \sqrt{2}(\mathcal{H}_{ba} + \mathcal{H}_{b\bar{a}}^c), \quad \text{for } |a\rangle = |\bar{a}\rangle, \text{ and } |b\rangle \neq |\bar{b}\rangle. \end{aligned} \quad (96)$$

Thus, just $\mathcal{H}$ acts on the singlets and the $\mathcal{H}^c$ -terms are given by the symmetry. Note that singlet-singlet transitions are impossible for $[\mathcal{H} + \mathcal{H}^c]_{\beta\alpha}$.

In conclusion, the normalization is $N_\alpha = \sqrt{2}$ for in-states and $M_j = \sqrt{2}$ for out-states, in the cases where either state is a singlet. Additional, there is the sign factor coming in, if the out-state is a secondary state.

All together can be written as

$$[\mathcal{H}^0 + \mathcal{H} + \mathcal{H}^c]_{\beta\alpha} = \sum_{j,i} \text{sgn}(M_j) D_{\beta j} \mathcal{H}_{ji}^0 C_{i\alpha} + \sum_{j,i} M_j D_{\beta j} \mathcal{H}_{ji} C_{i\alpha} N_\alpha + \sum_{\substack{j \\ |i\rangle \in \text{doublets}}} M_j D_{\beta j} \mathcal{H}_{ji}^c C_{i\alpha} \quad (97)$$

$$\text{with } N_\alpha = \begin{cases} \sqrt{2}, & |\alpha\rangle \text{ a singlet} \\ 1, & |\alpha\rangle \text{ a doublet} \end{cases} \quad \text{and} \quad M_j = \begin{cases} \sqrt{2}, & |j\rangle = |\bar{j}\rangle \\ 0, & |j\rangle = -|\bar{j}\rangle \\ 1, & |j\rangle \neq |\bar{j}\rangle, \text{ a reference state} \\ \langle j|(-1)^{\sum_k c_k^\dagger c_k}|j\rangle, & |j\rangle \neq |\bar{j}\rangle, \text{ a secondary state.} \end{cases} \quad (98)$$

The position of the matrix element is determined by $D_{\beta j} \mathcal{H}_{ji} C_{i\alpha}$, implemented as described in section 9.2. The amplitude gets two additional factors, where N_α is given by the first normalization factor in the C-basis eq.(89) and M_j is returned using the Association eq.(90).

For the example states, it is straight forward to show, that the Hamiltonian is

$$H_{\beta\alpha} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & (-1)\sqrt{2} \cdot 0.27 \\ 0 & 0 & 2 & 0 \\ 0 & (-1)\sqrt{2} \cdot 0.27 & 0 & 2.09 \end{pmatrix}, \quad (99)$$

and lead to the same eigenvalues as the full basis eq.(86) (excluding the charge odd states).

10 Diagrammatic Rules

The goal of this section is to calculate terms in the old fashioned perturbation theory[15]. The technique is needed to treat the UV divergence.

For example at second order one needs to calculate

$$\sum_{\alpha} \frac{\langle f|V|\alpha\rangle\langle\alpha|V|i\rangle}{E_i - E_{\alpha}}, \text{ with } V = \int d\theta g\phi\bar{\psi}\psi. \quad (100)$$

It will be shown that the brute force calculations can be put in a set of diagrammatic rules.

Moreover, the state- rather than the field representation is used for all states $|\alpha\rangle$. For the potential, the fields are kept as described (see eq.(32) and eq.(49)).

The idea is then as follows: The initial state may consist of an arbitrary particle $a^\dagger|0\rangle$, next to it, is the potential consisting of an arbitrary field φ . This can be written as

$$\langle 0|... \varphi a^\dagger ...|0\rangle = \overline{\varphi^-} a^\dagger \langle 0|...|0\rangle + \langle 0|... : \varphi^- a^\dagger : ...|0\rangle + \langle 0|... \varphi^+ a^\dagger ...|a\rangle \quad (101)$$

$$\Leftrightarrow \overline{AB} \equiv AB - :AB: \quad (102)$$

To prevent confusion, normal ordering notation is not put around expressions containing purely creation or annihilation operators. It is possible to proceed and write all fields and operators in terms of normal ordered expressions and Wick contractions plus additional terms, which contain either more creation or annihilation operators (like in the example). The additional terms give zero, because of the annihilation/creation operators acting on the vacuum. Wick contractions don't contain operators and can be factored out. Then, all terms with at least one normal ordered expressions left give zero, which leave only the fully contracted expressions.

Fermion fields contain vectors, and their operators anti-commute, thus the ordering is crucial. This problem can be treated systematically by keeping the fields in place, while commuting the operators from states around it. That way it is assured, that a co-vector $(\bar{\psi})$ will contract with the corresponding vector (ψ) .

Furthermore, the Wick contraction of fermions is defined with the specific ordering. In particular, the two operators must stand next to each other. Starting from the basis states (which are defined with an ordering, see eq.(68)), one needs to rearrange the operators first, which then defines the sign⁹. For example

$$\langle 0|d_1c_{-1}\phi\bar{\psi}\psi a_0^\dagger|0\rangle = \langle 0|d_1c_{-1}\phi\bar{\psi}\psi a_0^\dagger|0\rangle = (-1)^2(\overbrace{\phi a_0^\dagger}^{\overbrace{}})_0(\overbrace{c_{-1}\bar{\psi}}^{\overbrace{\phantom{c_{-1}\bar{\psi}}}}) \cdot (\overbrace{d_1\psi}^{\overbrace{}}).$$

This way each term in eq.(100) reduces to a sum over all possible, fully contracted expressions.

10.1 Vertex Factor

All three fields in V contain a factor $1/\sqrt{2\pi R e^{\pm i k \theta/R}}$. The integral over θ acts on these exponentials giving a delta function of momentum conservation times $2\pi R$ (see chapter 5.1). Hence each Vertex is

$$\begin{aligned} & \text{Diagram 1: A dashed line on the left splits into two solid lines on the right.} & = \frac{g}{\sqrt{2\pi R}} \delta(\sum k_{in} - \sum k_{out}) \\ \rightarrow & \text{Diagram 2: A dashed line on the left splits into two solid lines on the right. The top solid line is labeled 'n' and the bottom solid line is labeled '-m'.} & = \frac{g}{\sqrt{2\pi R}} \end{aligned} \quad (103)$$

⁹In time ordered perturbation theory diagrammatic rules include whether a diagram gets an additional sign or not. Here it turned out to be more handy to write the corresponding bra-ket notation of a diagram and reorder explicitly according to the Wick contractions.

Henceforth, the delta function is kept implicitly. Instead, particle lines must be written together with a momentum attached, which is an equivalent description.

In the diagrams momentum and time goes from right to left; dashed lines correspond to bosons, lines with a left pointing arrow to fermions and with a right pointing arrow to anti fermions.

10.2 Boson Lines

For the boson field (see eq.(32))

$$\phi = \sum_k \frac{1}{\sqrt{2\pi R}} (\phi_k^- e^{ik\theta/R} + \phi_k^+ e^{-ik\theta/R}) = \sum_k \frac{1}{\sqrt{2\pi R}} \left(\frac{a_k}{\sqrt{2\Omega_k}} e^{ik\theta/R} + \frac{a_k^\dagger}{\sqrt{2\Omega_k}} e^{-ik\theta/R} \right),$$

the Wick contraction with an incoming boson is

$$\overline{\phi} a_l^\dagger = \phi^- a_l^\dagger - a_l^\dagger \phi^- = \sum_k \frac{1}{\sqrt{2\pi R}} \frac{\delta_{k,l}}{\sqrt{2\Omega_k}} e^{-ik\theta/R} \rightarrow \frac{1}{\sqrt{2\Omega_l}}. \quad (104)$$

For the Feynman rules everything, already considered in the vertex factor must be omitted (this step is denoted as " $\rightarrow$ "). Analogous for the outgoing Boson:

$$\overline{a_l \phi} = a_l \phi^+ - \phi^+ a_l \rightarrow \frac{1}{\sqrt{2\Omega_l}}. \quad (105)$$

Contracting a field with a multiple particle state gives the same multiplicity factor as a creation/annihilation operator acting on a state. If $|n\rangle$ consists of n bosons with momentum l , then

$$\begin{aligned} \overline{\phi \frac{1}{\sqrt{n!}} (a_l^\dagger)^n |0\rangle} &= \overline{\phi} a_l^\dagger \sqrt{n} \left(\frac{1}{\sqrt{(n-1)!}} (a_l^\dagger)^{(n-1)} |0\rangle \right) \\ &\Leftrightarrow \overline{\phi |n\rangle} = \sum_l \overline{\phi} a_l^\dagger \overline{a_l |n\rangle}, \end{aligned} \quad (106)$$

$$\text{analogous} \quad \langle n | \phi = \sum_l \langle n | a_l^\dagger \overline{a_l \phi} \quad (107)$$

Although, the l.h.s. are more frequently used, the r.h.s. are closer to the implementation, since the momentum appears explicitly.

Then, one can define the propagator as a field from the right potential in eq.(100), contracted with the intermediate state, multiplied by the left potential contracted with the same particle. This gives

$$\begin{aligned} \sum_l \dots \phi \dots a_l^\dagger \dots |0\rangle \langle 0| \dots a_l \dots \phi &= \sum_l \overline{\phi} a_l^\dagger \overline{a_l \phi} (\dots |0\rangle \langle 0| \dots) + \text{other contractions} \\ \sum_l \overline{\phi} a_l^\dagger \overline{a_l \phi} &= \frac{1}{2\pi R} \sum_{k_1, k_2} \frac{\delta_{k_1, k_2}}{2\Omega_{k_1}} e^{i(k_1 - k_2)\theta/R} \rightarrow \frac{1}{2\Omega_l}. \end{aligned} \quad (108)$$

In order to use these calculations as diagrammatic rules, one must not consider multiplicity for intermediate states. This is shown in the appendix 18.

Multiparticle Feynman rules with explicit momentum are

$$\bullet \text{---} \overset{k}{\text{---}} = \frac{a_k}{\sqrt{2\Omega_k}} \quad (109)$$

$$\text{---} \overset{k}{\text{---}} \bullet = \frac{a_k^\dagger}{\sqrt{2\Omega_k}} \quad (110)$$

$$\bullet \text{---} \overset{k}{\text{---}} \bullet = \frac{1}{2\Omega_k} \quad (111)$$

10.3 Fermion Lines

The fermion fields are (see eq.(49))

$$\begin{aligned} \psi &= \sum_k \frac{1}{\sqrt{2\pi R}} (\psi_k^- e^{ik\theta/R} + \psi_k^+ e^{-ik\theta/R}) = \sum_k \frac{1}{\sqrt{2\pi R}} \left(\mathbf{v}_k \frac{c_k}{\sqrt{2\omega_k}} e^{ik\theta/R} + \mathbf{v}_k^* \frac{d^\dagger}{\sqrt{2\omega_k}} e^{-ik\theta/R} \right) \\ \bar{\psi} &= \sum_k \frac{1}{\sqrt{2\pi R}} (\bar{\psi}_k e^{ik\theta/R} + \bar{\psi}_k^+ e^{-ik\theta/R}) = \sum_k \frac{1}{\sqrt{2\pi R}} \left(\mathbf{v}_k^\dagger \gamma^0 \frac{c_k^\dagger}{\sqrt{2\omega_k}} e^{ik\theta/R} + \mathbf{v}_k^T \gamma^0 \frac{d}{\sqrt{2\omega_k}} e^{-ik\theta/R} \right), \\ \text{with } \mathbf{v}_k^T &= (\sqrt{\omega_k + \frac{k}{R}}, i\sqrt{\omega_k - \frac{k}{R}}). \end{aligned}$$

Normal ordering creates a minus sign for fermion operators. That is why one needs to calculate the anticommutation relation rather than the commutation relation for the Wick contractions. This gives

$$\overline{\psi} c_k^\dagger = \{\psi^-, c_k^\dagger\} \rightarrow \frac{\mathbf{v}_k}{\sqrt{2\omega_k}} \quad (112)$$

$$\overline{d}_k \psi = \{d_k, \psi^+\} \rightarrow \frac{\mathbf{v}_k^*}{\sqrt{2\omega_k}} \quad (113)$$

$$\overline{\psi} d_k^\dagger = \{\bar{\psi}, d_k^\dagger\} \rightarrow \frac{\bar{\mathbf{v}}_k}{\sqrt{2\omega_k}} \quad (114)$$

$$c_k \bar{\psi} = \{c_k, \bar{\psi}^+\} \rightarrow \frac{\bar{\mathbf{v}}_k}{\sqrt{2\omega_k}}, \quad (115)$$

The fermion propagator is associated with the configuration

$$\langle 0 | \dots \overline{\psi} \psi \dots c_k^\dagger \dots | 0 \rangle \langle 0 | \dots \overline{c}_k \dots \bar{\psi} \psi \dots | 0 \rangle \rightarrow \left[\overline{\psi} c_k^\dagger \otimes \overline{c}_k \bar{\psi} \right] = \frac{1}{2\omega_k} [\mathbf{v}_k \otimes \bar{\mathbf{v}}_k]. \quad (116)$$

In the context of spinors, squared brackets are used to emphasize, that an expression must act from the right on spinors or from the left on adjoined spinors.

To identify the anti fermion propagators in the fully contracted expressions, it is used that $\bar{\psi}\psi$ forms a scalar as soon as the fields are contracted with two fermions. In this case, they can be reordered. The anti fermion propagator is

$$\langle 0 | \dots \overline{\psi} \psi \dots d_k^\dagger \dots | 0 \rangle \langle 0 | \dots \overline{d}_k \dots \bar{\psi} \psi \dots | 0 \rangle \rightarrow \left[\overline{d}_k \psi \otimes \bar{\psi} d_k^\dagger \right] = \frac{1}{2\omega_k} [\mathbf{v}_k^* \otimes \bar{\mathbf{v}}_k^*]. \quad (117)$$

Calculating the propagators explicitly gives

$$\mathbf{v}_k \otimes \mathbf{v}_k^\dagger \gamma^0 = \begin{pmatrix} \sqrt{\omega_k + \frac{k}{R}} \\ i\sqrt{\omega_k - \frac{k}{R}} \end{pmatrix} \otimes \begin{pmatrix} \sqrt{\omega_k - \frac{k}{R}} & -i\sqrt{\omega_k + \frac{k}{R}} \end{pmatrix} = \begin{pmatrix} m_\psi & -i(\omega_k + \frac{k}{R}) \\ i(\omega_k - \frac{k}{R}) & m_\psi \end{pmatrix}$$

$$\Rightarrow \mathbf{v}_k \otimes \overline{\mathbf{v}_k} = \not{k} + m_\psi \quad (118)$$

$$\mathbf{v}_k^* \otimes \overline{\mathbf{v}_k^*} = -(\mathbf{v}_k \otimes \overline{\mathbf{v}_k})^* = \not{k} - m_\psi, \quad (119)$$

where the minus in the last line comes from complex conjugating γ^0 .

As for the boson case, the multiplicity for intermediate states can be ignored, i.e. the Pauli exclusion principle for intermediate states does not apply. This is expected, otherwise loops would depend on spectator particles. An explicit derivation can be found in appendix 18

This concludes the Feynman rules

$$\begin{array}{c} \text{---} \overleftarrow{k} \text{---} \\ \text{---} \overrightarrow{k} \text{---} \end{array} = \frac{\mathbf{v}_k}{\sqrt{2\omega_k}} c_k \quad \begin{array}{c} \text{---} \overleftarrow{k} \text{---} \\ \text{---} \overrightarrow{k} \text{---} \end{array} = \frac{\overline{\mathbf{v}_k^*}}{\sqrt{2\omega_k}} d_k \quad (120)$$

$$\begin{array}{c} \text{---} \overleftarrow{k} \text{---} \\ \text{---} \overrightarrow{k} \text{---} \end{array} = \frac{\overline{\mathbf{v}_k}}{\sqrt{2\omega_k}} c_k^\dagger \quad \begin{array}{c} \text{---} \overleftarrow{k} \text{---} \\ \text{---} \overrightarrow{k} \text{---} \end{array} = \frac{\mathbf{v}_k^*}{\sqrt{2\omega_k}} d_k^\dagger \quad (121)$$

$$\begin{array}{c} \text{---} \overleftarrow{k} \text{---} \\ \text{---} \overrightarrow{k} \text{---} \end{array} = \frac{[\not{k} + m_\psi]}{2\omega_k} \quad \begin{array}{c} \text{---} \overleftarrow{k} \text{---} \\ \text{---} \overrightarrow{k} \text{---} \end{array} = \frac{[\not{k} - m_\psi]}{2\omega_k}. \quad (122)$$

10.4 Simplified Feynman Rules

The usual Feynman rules are handy to describe physics (see e.g.[12][13]), but they have the disadvantage, that one has to deal with matrix and vector multiplication after applying them. For old fashioned perturbation theory the contracted terms can be reordered such that only scalars appear in diagrammatic rules.

For the derivation momentum is attached to the Wick contractions as in eq.(106) or eq.(107). Similar, propagators are written as

$$\overline{\varphi}[\alpha] \langle \alpha | \varphi = \overline{\varphi} b_k^\dagger \overline{b_k} b_l^\dagger \overline{b_l} \varphi. \quad (123)$$

It is used, that multiplicity for intermediate states doesn't apply.

Then, the operators and the magnitude can be separated. For example a term in the expansion at 2nd order eq.(100) could be Wick contracted as

$$\overline{\langle f | \phi \psi \psi | \alpha \rangle} \langle \alpha | \phi \psi \psi | i \rangle = \overline{\langle f | c_n^\dagger \left(\overline{\phi} a_{n-k}^\dagger \cdot \overline{c_n} \overline{\psi} \cdot \overline{\psi} d_k^\dagger \right) a_{n-k} \overline{d_k} d_l^\dagger a_{m-l}^\dagger \left(a_{m-l} \overline{\phi} \cdot \overline{d_l} \overline{\psi} \cdot \overline{\psi} c_m^\dagger \right) c_m | i \rangle}$$

The contraction outside the parentheses are defined already. Especially, those in the middle give delta functions, which can be omitted by considering momentum conservation, when drawing diagrams. The

ones inside form a scalar, which is used as "contracted vertex factor". The advantage is, that there are only two types of these. The new Vertex factors are

$$\begin{array}{c} \text{---} n-m \text{---} \\ \swarrow \quad \searrow \\ n \quad -m \end{array} = \frac{g}{\sqrt{2\pi R}} \left(\overline{a_{n-m} \phi \psi d_n^\dagger \psi c_{-m}^\dagger} \right) a_{n-m}^\dagger d_n c_{-m} = \frac{g \operatorname{sgn}(n+m)}{2\sqrt{\pi R \Omega_{n-m}}} \sqrt{\frac{1}{2} - M_{nm}} a_{n-m}^\dagger d_n c_{-m} \quad (124)$$

$$\begin{array}{c} n \\ \swarrow \quad \searrow \\ \text{---} n-m \text{---} \\ -m \end{array} = \frac{g}{\sqrt{2\pi R}} \left(\overline{\phi a_{n-m}^\dagger c_{-m} \psi d_n^\dagger \psi} \right) a_{n-m} c_{-m}^\dagger d_n^\dagger = \frac{g \operatorname{sgn}(n+m)}{2\sqrt{\pi R \Omega_{n-m}}} \sqrt{\frac{1}{2} - M_{nm}} a_{n-m} c_{-m}^\dagger d_n^\dagger \quad (125)$$

$$\begin{array}{c} m \\ \swarrow \quad \searrow \\ n \text{---} \\ \text{---} n-m \end{array} = \frac{g}{\sqrt{2\pi R}} \left(\overline{a_{n-m} \phi c_m \psi \psi c_n^\dagger} \right) a_{n-m}^\dagger c_m^\dagger c_n = \frac{g}{2\sqrt{\pi R \Omega_{n-m}}} \sqrt{\frac{1}{2} + M_{nm}} a_{n-m}^\dagger c_m^\dagger c_n \quad (126)$$

$$\begin{array}{c} m \\ \swarrow \quad \searrow \\ n \text{---} \\ \text{---} n-m \end{array} = \frac{g}{\sqrt{2\pi R}} \left(\overline{a_{n-m} \phi \cdot \psi d_n^\dagger \cdot d_m \psi} \right) a_{n-m}^\dagger d_n d_m^\dagger = \frac{g}{2\sqrt{\pi R \Omega_{n-m}}} \sqrt{\frac{1}{2} + M_{nm}} a_{n-m}^\dagger d_m^\dagger d_n. \quad (127)$$

It is with no surprise, that these factors are the same as the three particle interactions calculated in eq.(60). The boson could either be outgoing or incoming, the factor stays the same. So, these diagrams represent all cases.

Note that, the first two terms are defined in a way, such that the free momentum of fermions needs to be written with a minus sign (which is no more then a convention).

This concludes the diagrammatic rules:

1. Draw the Feynman diagram.
2. Attach momentum variables to each line, such that:
 - Momentum is conserved.
 - Anti-fermion momentum is defined with a plus sign.
 - Fermion momentum is defined with a minus sign.
3. Each Vertex with a (anti-)fermion passing by get a factor of

$$\frac{g}{2\sqrt{\pi R\Omega_{n-m}}} \sqrt{\frac{1}{2} + M_{nm}},$$

where n and m are the momenta of the (anti-)fermions.

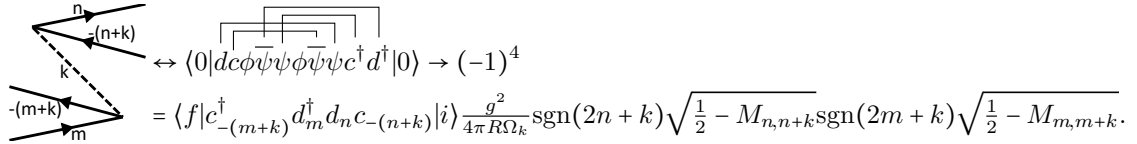
4. Each Vertex describing an annihilation or creation of fermions get a factor of (128)

$$\frac{g \operatorname{sgn}(n+m)}{2\sqrt{\pi R\Omega_{n-m}}} \sqrt{\frac{1}{2} - M_{nm}},$$

where n and m are the momenta of the anti-fermion and fermion, but the fermion momentum enters with a minus sign (see rule 2).

5. Each external leg is described with the creation/annihilation operator associated with the particle line. Those act on the external states. The fermion operators are properly ordered (see eq.(68)).
6. An additional sign factor is determined by using conventional Wick contractions i.e (anti-)commute the operators to the fields, specified by the Wick contraction.

One way to determine the sign of the last rule is to write down a diagram in it's field description and the relevant operators of the external states. Then put the Wick contractions as defined by the diagram and count, how often fermionic contractions cross. The sign is (-1) to the power of crossings. For example,



This is one possible contraction of $\langle f|V|\alpha\rangle\langle\alpha|V|i\rangle$. If used in an expression e.g. eq.(100), one needs to divide by a factor like $E_i - E_\alpha$ given by the mathematical expression.

11 Hamiltonian Truncation Effective Theory

Hamiltonian Truncation Effective Theory (HTET)[1] will be applied to improve the convergence for increasing truncation energy E_{max} .

Moreover, this theory posses a UV divergence, hence renormalization must take place as well. As shown by Delouche et al.[2] both concepts are linked and will be applied to the Yukawa model.

11.1 Power Counting

Local Approximation:

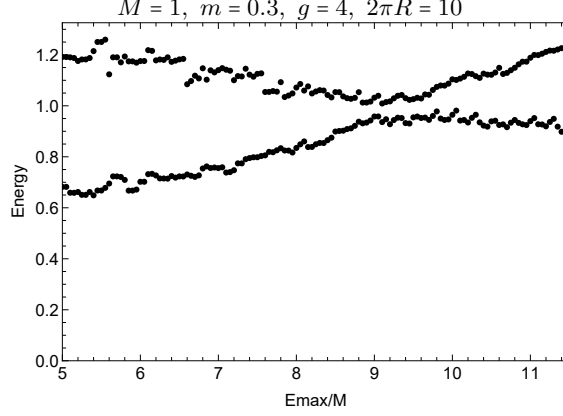


Figure 1: Shown are the differences of the 2nd and 3rd eigenvalues to the lowest one. This is then the physical energy of the excited states. Used is the unmodified Hamiltonian for different truncation energies E_{max} , with boson mass $M = 1$, fermion mass $m = 0.3$, volume $R = 10/(2\pi)$ and coupling $g = 4$. The data point don't converge, but depend on E_{max} , which indicates that renormalization has to be applied.

The method of power counting is presented in [1]. The important quantities are g with $\dim=1$, ϕ with $\dim=0$ and $\bar{\psi}\psi$ with $\dim=1$. Since $\mathcal{L}$ has $\dim=2$, the large scale behaviour of loops is expected to be

$$\mathcal{L} \propto g^2 \int d^2x \left[\mathbb{1} \left(\ln(E_{max}) + \frac{H_0}{E_{max}} \right) + \phi^2 \left(\ln(E_{max}) + \frac{H_0}{E_{max}} \right) + \bar{\psi}\psi \frac{1}{E_{max}} + o(1/E_{max}^2) \right] \rightarrow H_2. \quad (129)$$

The fields in this description represent external legs and the power of g give the number of vertices in loops. Every term is also expanded in a local expansion, which is given in powers of $1/E_{max}$. The terms, which aren't included produce an error, which vanish as $1/E_{max}^2$.

In truncation $E_{max} = \Lambda$ acts as natural cutoff. Power counting shows, that the eigenvalues do not converge and the deviation grows logarithmically.

The truncated Hamiltonian was implemented as described in section 9. The result for varying E_{max} is displayed in figure 1. For exemplary parameter, in particular a coupling of $g = 4$, the logarithmic divergence can not be identified clearly. Nevertheless, it is evident, that the energy levels do not converge.

In order to treat the divergence, it is sufficient to take the effective Hamiltonian and the counter terms at leading order. Taking all small scale quantities to zero give

$$H_2|_{E/E_{max}=0} \sim g^2 \int d^2x \left[\phi^2 \ln(E_{max}) + o(1/E_{max}) \right]. \quad (130)$$

The error vanish as $1/E_{max}$. Although, the vacuum itself diverges, in the difference of energy, it gives a $1/E_{max}$ distribution (see next chapter).

Exact Calculations:

The above argumentation follows strictly the systematics of [1]. The method has proven to be valid and is to be detached from the following argumentation. In the next chapters, it will be tested, if the effective Hamiltonian can be modified. The idea is to leave the expansion in g , but drop the local approximation.

For a effective theory in general, the eigenvalue spectrum of a Hamiltonian $\Lambda(H_\infty)$, where the truncation energy goes to infinity, is equivalent to $\Lambda(H_{eff})$ in the truncated Fock space. H_{eff} is given

in an infinite expansion of the coupling. In this description only a finite part can be calculated. For example at order two, H_{eff} is given in the form

$$\begin{aligned} H_{eff} &= H_0 + H_1 + H_2 + H_3 + \dots \\ \rightarrow H_{eff}^{(2)} &\equiv H_0 + H_1 + H_2 \\ H_{eff}^{(>2)} &= H_3 + H_4 \dots \end{aligned}$$

Taking a finite part of H_{eff} leave a error. The magnitude can be estimated. For example, one can show that

$$\begin{aligned} \delta &= \|\Lambda(H_\infty) - \Lambda(H_{eff}^{(2)})\| = \|\Lambda(H_{eff}) - \Lambda(H_{eff}^{(2)})\| \\ \rightarrow \delta &= o(\|H_{eff}^{(>2)}\|) = o\left(\frac{1}{E_{max}^2}\right). \end{aligned}$$

This demonstrate that the deviation of the eigenvalues is coming from the terms, which are not considered. The not included parts have summands with different $(1/E_{max})^n$ large scale behaviour. For large E_{max} the leading power of n will indicate the convergence of the eigenvalues for varying E_{max} .

Note that, if multiple not included parts dictate the large scale behaviour, it should not matter, if all of these terms or at least one are not included. The diagonalization mixes states, such that the convergence could just as well be given by one missing term.

A simple way to derive the scaling with E_{max} for different terms, is to use the local approximation as described by Cohen et al.[1]. Afterwards, it should be possible to resum all terms of the expansion. In eq.(129) the terms in parenthesis can be merged to

$$H_2 \sim g^2 \int d^2x \left[\mathbb{1} \ln(E_{max}) + \phi^2 \ln(E_{max}) + \bar{\psi} \psi \frac{1}{E_{max}} + o(1/E_{max}^2) \right]. \quad (131)$$

The smallest power of $1/E_{max}$ is given by the leading order in the local approximation of each term. The higher powers should die off for large energies.

The sum of all terms in the approximation is equivalent to not apply the expansion at first place. The associated power is given by the leading order, where all small scale quantities are zero as $1/E_{max} = 0$. So,

$$H_n = o(\|H_n|_{E/E_{max}=0}\|) + \text{higher order terms}. \quad (132)$$

In this argumentation, the error part of H_{eff} is short of some higher order terms, which are resummed. This should not change the convergence behaviour, if at least one term with the leading order of $1/E_{max}$ is left. Here, this is given by the terms

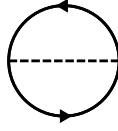
$$H_3 \sim g^3 \int d^2x \left[+\phi^3 \left(\frac{1}{E_{max}}\right)^3 + \phi \bar{\psi} \psi \left(\frac{1}{E_{max}}\right)^2 \right].$$

In this way, a $1/E_{max}^2$ convergence is expected for H_2 , just as for the expansion eq.(129).

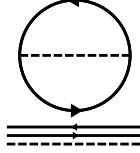
In the following chapters the exact effective Hamiltonian is derived. Then the leading order of the local approximation is taken. In section 12 both methods are compared.

11.2 Vacuum Renormalization

The vacuum is used to demonstrate the renormalization scheme and to introduce the effective Hamiltonian. The Feynman diagram



has a logarithmic UV divergence. In a full theory this diagram gets cancelled in the end, because to all transitions also the disconnected diagram



contributes. The loop factorizes, such that the amplitude is the same as the vacuum loop. In the difference between any two energy levels, these diagrams are subtracted against each other. On the other side, in truncation the energy of intermediate states must stay below E_{max} . This also depend on spectator particles. If they have energy E , the loop energy goes up to $E_{max} - E$. Because the amplitude grows logarithmically, after subtraction of the vacuum, the behaviour is

$$V \propto g^2 (\ln(E_{max} - E) - \ln(E_{max})) = -\frac{g^2 E}{E_{max}} \left(\frac{\ln 1 - \ln(1 - E/E_{max})}{E/E_{max}} \right) \xrightarrow{E_{max} \gg E} -\frac{g^2 E}{E_{max}}. \quad (133)$$

In the parenthesis $\ln 1 = 0$ is added, such that the derivative $\frac{\partial \ln x}{\partial x} \Big|_1 = 1$ is formed.

Similar, one can show, that the power of $1/E_{max}$ in vacuum diagrams is always raised by one, if the difference of energies are taken¹⁰.

To renormalize the vacuum, it is always possible to add a scalar term to the Hamiltonian. Including the effective Hamiltonian, the term to calculate will be

$$H = H_0 + H_{eff} + \delta_V. \quad (134)$$

In the matrix description, δ_V changes the eigenvalue equation to

$$|H + \mathbb{1}\delta_V - \mathbb{1}\lambda| = 0.$$

It shifts all solutions by δ_V , which didn't changes the difference between eigenvalues. δ_V is therefore a free parameter, which can be chosen, such that the vacuum is renormalized.

¹⁰The general derivation for an amplitude $f(E)$, which is evaluated at $E - E_{max}$ for a exited state, minus the amplitude evaluated at E_{max} for the vacuum, is

$$\begin{aligned} -f(E_{max}) + f(E_{max} - E) = & -\frac{E}{E_{max}} \left(\frac{f(E_{max} \cdot 1) - f(E_{max}[1 - E/E_{max}])}{E/E_{max}} \right) \\ & \xrightarrow{E_{max} \gg E} -\frac{E}{E_{max}} \frac{\partial}{\partial x} f(E_{max} \cdot x) \Big|_{x=1}. \end{aligned}$$

Since x has now dimension, the power of $1/E_{max}$ is raised by one.

The effective Hamiltonian H_{eff} is written in an expansion of the potential[1]. Up to second order, the terms of interest are¹¹

$$\begin{aligned} H_{eff} &= H_1 + H_2 + \dots \\ \langle f|H_1|i\rangle &= \langle f|V|i\rangle \end{aligned} \quad (135)$$

$$\langle f|H_2|i\rangle = \sum_{\alpha}^{\geq} \frac{\langle f|V|\alpha\rangle\langle\alpha|V|i\rangle}{E_i - E_{\alpha}}, \quad (136)$$

where, the sum is defined as

$$\sum_{\alpha}^{\geq} \equiv \sum_{\alpha}^{\geq} \theta(E_{\alpha} - E_{max}) \quad (137)$$

$$\text{with } \overset{\geq}{\theta}(E) = \begin{cases} 1, & E > 0 \\ 0, & \text{else,} \end{cases} \quad \text{or } \overset{\leq}{\theta}(E) = \begin{cases} 1, & E \leq 0 \\ 0, & \text{else,} \end{cases} \dots \quad (138)$$

Moreover, the effective Hamiltonian is split into

$$H_2 = H_{2,V} + H_{2,M} + H_{2,m},$$

Where $H_{2,V}$ is related to the vacuum diagram, $H_{2,M}$ to boson corrections and $H_{2,m}$ to fermion corrections.

11.2.1 Vacuum Counter Term

The amplitude for the disconnected 2nd order vacuum loop, using old fashioned perturbation theory, is

$$\begin{aligned} \int d\theta g^2 \sum_{\alpha} \frac{\langle f|\phi\bar{\psi}\psi|\alpha\rangle\langle\alpha|\phi\bar{\psi}\psi|i\rangle}{E_i - E_{\alpha}} \supset \text{Diagram} &\leftrightarrow \langle f|\phi\bar{\psi}\psi\phi\bar{\psi}\psi|i\rangle \rightarrow (-1)^0 \\ &= \frac{-g^2}{2\pi R} \sum_{n,k} \frac{\frac{1}{2} - M_{k-n/2,k+k-n/2}}{2\Omega_n(\Omega_n + \omega_{k-n/2} + \omega_{k+n/2})} \langle f|i\rangle \\ &= \frac{-g^2}{2\pi R} \sum_{n,k} \frac{\omega_{k-n/2}\omega_{k+n/2} - m^2 + (k^2 - n^2/4)/R^2}{4\Omega_n\omega_{k-n/2}\omega_{k+n/2}(\Omega_n + \omega_{k-n/2} + \omega_{k+n/2})} \langle f|i\rangle, \end{aligned} \quad (139)$$

Where the expression is written down using eq.(128). The kinematic variables are chosen, such that the expression is symmetric under $n \rightarrow -n$ and $k \rightarrow -k$. Using the symmetries speed up the calculations by a factor of 4. On the other hand k takes integers or half integers

$$\{k + n/2\} \in \mathbb{Z} \rightarrow \begin{cases} \{k\} \in \mathbb{Z}, & \text{if } n \text{ even} \\ \{k + 1/2\} \in \mathbb{Z}, & \text{if } n \text{ odd.} \end{cases}$$

The dimension of the summand is -2, while the sum raises the dimension by 2. If one introduces a cutoff, the term diverges indeed logarithmically.

¹¹The eigenvalues of a matrix H are invariant under the transformation $H \rightarrow H^T$, or equivalently $\langle f|H|i\rangle \rightarrow \langle i|H|f\rangle$. The property is used to change $E_f \rightarrow E_i$ in contrast to the original description. That makes the terms more convenient for the implementation.

δ_V is then chosen to counteract this divergence. Since eq.(139) is in the form of a scalar, one can directly set

$$\delta_V = \frac{g^2}{2\pi R} \sum_{n,k} \frac{\frac{1}{2} - M_{k-n/2, k+n/2}}{2\Omega_n(\Omega_n + \omega_{k-n/2} + \omega_{k+n/2})} \quad (140)$$

11.2.2 Vacuum Effective Hamiltonian

In the language of Feynman diagrams the counter term is chosen, such that

$$\text{---} \text{---} \text{---} + \text{---} \text{---} \text{---} + \text{---} \text{---} \text{---} = \text{---} \text{---} \text{---},$$

for arbitrary spectator particles.

This seems to be satisfied, since the counter term is exactly the negative of the loop, but the energy in the loop of the counter term runs up to infinity, while the bare diagram runs up to E_{max} in truncation. Therefore, $H_{2,V}$ must be added, such that, the loop runs effectively up to infinity. In this way counter terms can only be added together with an effective Hamiltonian, as described by Delouche et al.[2]. Roughly speaking, the effective Hamiltonian enlarges particular Feynman diagrams above the truncation limit.

The calculations are as before, just the Heaviside function is added. The effective Hamiltonian of the vacuum $H_{2,V}$ is

$$H_{2,V} \subset H_2 = \sum_{\alpha} \frac{\langle f|V|\alpha\rangle\langle\alpha|V|i\rangle}{E_i - E_{\alpha}}$$

$$\langle f|H_{2,V}|i\rangle = \frac{-g^2}{2\pi R} \sum_{n,k} \frac{\frac{1}{2} - M_{k-n/2, k+n/2}}{2\Omega_n(\Omega_n + \omega_{k-n/2} + \omega_{k+n/2})} \langle f|i\rangle^{\geq} \theta(E_i + \Omega_n + \omega_{k-n/2} + \omega_{k+n/2} - E_{max}). \quad (141)$$

By renormalizing the vacuum, the Hamiltonian becomes

$$H = H_0 + H_1 + (H_{2,V} + \delta_V),$$

$$\text{with } \langle f|H_{2,V} + \delta_V|i\rangle = \frac{g^2}{2\pi R} \sum_{n,k} \frac{\frac{1}{2} - M_{k-n/2, k+n/2}}{2\Omega_n(\Omega_n + \omega_{k-n/2} + \omega_{k+n/2})} \langle f|i\rangle^{\leq} \theta(\Omega_n + \omega_{k-n/2} + \omega_{k+n/2} + E_i - E_{max}). \quad (142)$$

The counter term and the effective Hamiltonian are added together, using $1 - \overset{\geq}{\theta}(E) = \overset{\leq}{\theta}(E)$.

The renormalization is tested, by calculating the eigenvalues for varying small couplings. In this regime perturbation theory is valid. Without renormalization the leading order $o(g^2)$ diagrams are calculated. This order should be cancelled with the renormalization, such that the next to leading order $o(g^4)$ is most significant. In figure 2 both methods are displayed.

For example parameters $M = 1$, $m = 0.3$, $R = 10/\sqrt{2\pi}$, $E_{max} = 8$ and coupling up to $g = 1.5$, the graphics show the expected behaviour of the perturbative regime, i.e. the measure points can be described with

$$E(g) = a_2 g^2 + a_4 g^4 + a_6 g^6$$

The second order term $o(g^2)$ is indeed suppressed with magnitude of $a_2 = -0.008 \pm 0.004$, while the fourth order coefficient is $a_4 = -0.012 \pm 0.005$ and the $o(g^2)$ factor of the unmodified Hamiltonian is $a_2^{(0)} = -0.518 \pm 0.004$.

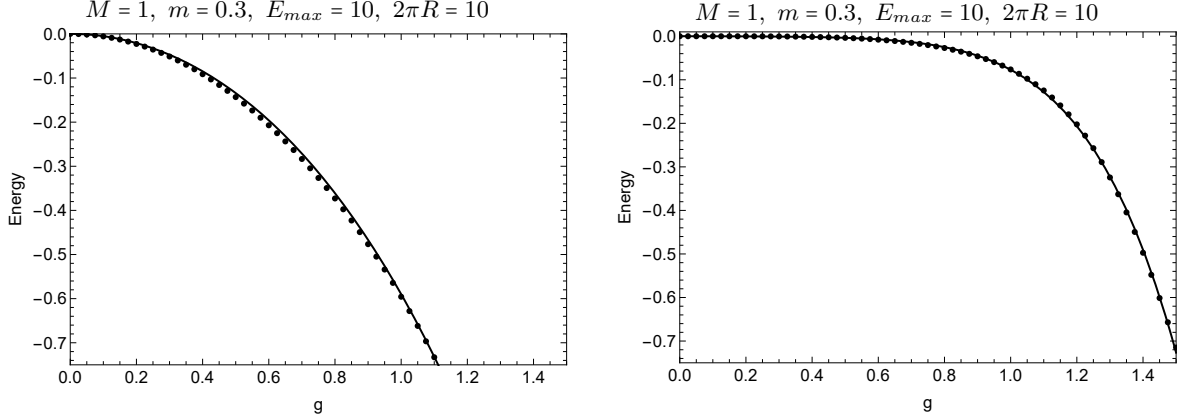


Figure 2: On the l.h.s. are the lowest eigenvalues of the unmodified Yukawa Hamiltonian shown, while on the r.h.s. the vacuum correction is added. The used parameter are $M = 1$, $m = 0.3$, $R = 10/\sqrt{2\pi}$ and $E_{max} = 10$. In the coupling range perturbation theory is at hand, which is tested, by fitting each plot up to next to leading order. On the l.h.s the fit function is $E(g) = -0.518g^2 - 0.071g^4$ on the r.h.s. it is $E(g) = -0.008g^2 - 0.012g^4 - 0.057g^6$. As expected, the second order term is suppressed by the vacuum correction with coefficient -0.008 ± 0.004 , which is smaller than the coefficient -0.518 ± 0.004 of the unmodified Hamiltonian.

Since

$$a_2 < a_4 \ll a_2^{(0)},$$

the renormalization is proven to be valid.

11.3 Mass Terms

For other renormalizations, it needs to be explored which freedom the theory posses to add additional terms.

The boson loop with two external boson legs is diverging logarithmically. To counter the diagram, it is used, that the mass terms in the Lagrangian can be split, such that

$$\mathcal{L} = \frac{1}{2}(\partial\phi)^2 - \frac{1}{2}\overline{M}^2\phi^2 - \frac{1}{2}\delta M^2\phi^2 + \overline{\psi}(i\cancel{D} - \overline{m})\psi - \delta m\overline{\psi}\psi + \mathcal{L}^{int} \quad (143)$$

with $\overline{M}^2 + \delta M^2 = M^2$ and $\overline{m} + \delta m = m$.

The extra terms δM and δm are treated as interaction, while $\overline{M}$ and $\overline{m}$ terms contribute to the free Hamiltonian.

If, the coupling is set to zero, the eigenvalues needs to converge to the original masses for $E_{max} \gg M, m$. This way, the theory can be tested. Thereafter, the boson divergent diagram is put in the form of $\frac{1}{2}\delta M^2\phi^2$ in order to show the contribution to the mass.

11.3.1 Boson Mass Term

The scalar field eq.(32) can be directly inserted. Note, that the free energy is now

$$\Omega_n = \sqrt{\overline{M}^2 + \left(\frac{n}{R}\right)^2}.$$

The Hamiltonian gets an additional term of

$$\begin{aligned}
V_{\delta M} &= - \int d\theta \left(-\frac{1}{2} \delta M^2 \right) : \phi \phi : \\
&= \frac{\delta M^2}{2} \sum_{k,l} \int \frac{d\theta}{2\pi R \sqrt{4\Omega_k \Omega_l}} : [a_k^\dagger e^{-ik\theta/R} + a_k e^{ik\theta/R}] [a_l^\dagger e^{-il\theta/R} + a_l e^{il\theta/R}] : \\
\Leftrightarrow V_{\delta M} &= \sum_k \frac{\delta M^2}{2\Omega_k} (a_k^\dagger a_k + \frac{1}{2} a_k^\dagger a_{-k}^\dagger + \frac{1}{2} a_k a_{-k})
\end{aligned} \tag{144}$$

As described in section 5.1 the time and energy dependent exponentials can be omitted.

There is a difference to the unmodified theory, coming from the vacuum loop. Power counting yields

$$\mathcal{L} \propto \int d^2 x \left[\mathbb{1} \frac{\delta M^4}{E_{max}^2} + o(1/E_{max}^3) \right]. \tag{145}$$

As described in the discussion of the vacuum, the error is raised by one power in the difference of energies. The expected behaviour of energy levels is

$$E(E_{max}) = a + b \left(\frac{1}{E_{max}} \right)^3, \tag{146}$$

where a should equal the free energy in the unmodified theory H_0 .

Implementing the free Hamiltonian and the boson mass term, where the mass term is split in half, so $\overline{M}^2 = \delta M^2 = 0.5 M^2$ is shown in figure 3. The model converge to the expected values. The difference of the individual limits to the proper values are of order $E_{dif}/E_0 = o(0.0001)$.

Therefore, splitting the boson mass is equivalent to the free theory.

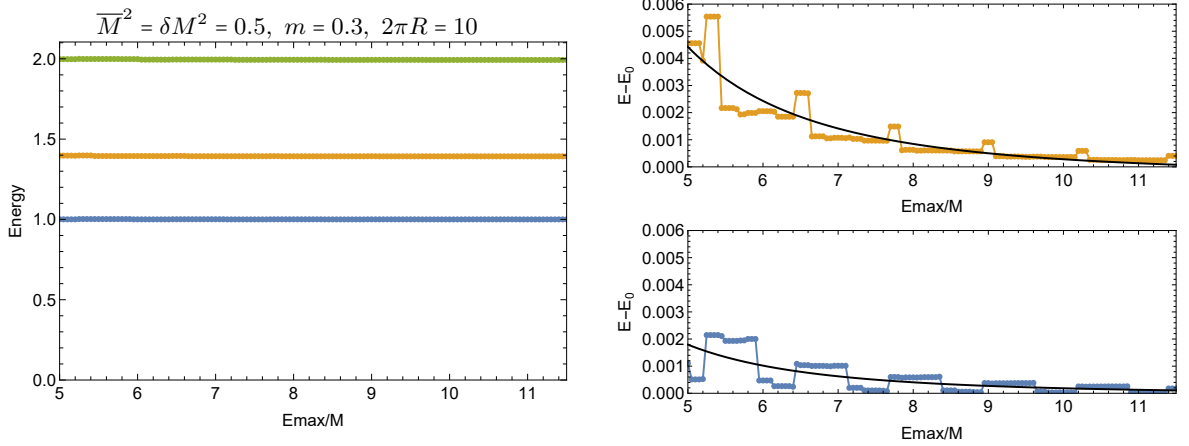


Figure 3: Shown on the l.h.s are the lowest eigenvalues of the free Hamiltonian and additional boson mass term.

The parameter are $R = 10/2\pi$, $m = 0.3$ and $\overline{M}^2 = \delta M^2 = 0.5 M^2 = 0.5$ and the vacuum is subtracted. $\overline{M}^2$ contributes to the free Hamiltonian and δM is added as interaction. On the r.h.s. are the difference between the data point from the lowest two eigenvalues and the expected values, i.e. the free energies of the lowest basis states. The data points are connected to emphasize the discontinuous behaviour. The used fit function is $a + b(1/x)^3$, where a states the convergence to the original free Hamiltonian. The 1st level converge to $a_1 = -0.00004 \pm 0.00006$ below the expected value of $E_1 = 1$ with $b_1 \simeq 0.23$ and the 2nd to $a_2 = -0.00031 \pm 0.00007$ above $2\sqrt{0.3^2 + R^{-2}} = 1.39$ with $b_2 \simeq 0.59$. Therefore, both models are equivalent

11.3.2 Fermion Mass Term

Although, the fermion will not be renormalized, the same method is applied, in order to test the model. It is straight forward to calculate¹² that the fermion mass term is given as

$$\begin{aligned}
V_{\delta m} &= -i\delta m \int d\theta (\psi_1^\dagger \psi_2 - \psi_2^\dagger \psi_1) \\
&= \int d\theta \sum_{k,l} \frac{-i\delta m}{2\pi R \sqrt{4\omega_k \omega_l}} [v_{1,k}^* c_k^\dagger e^{-ik\theta/R} + v_{1,k} d_k e^{ik\theta/R}] [v_{2,l} c_l e^{il\theta/R} + v_{2,l}^* d_l^\dagger e^{-il\theta/R}] - (1 \leftrightarrow 2) \\
&= \sum_k \frac{-i\delta m}{2\omega_k} [2i\bar{m} (c_k^\dagger c_k + d_k^\dagger d_k) + 2\frac{k}{R} (c_{-k}^\dagger d_k^\dagger + d_k c_{-k})] \\
&\Leftrightarrow V_{\delta m} = \sum_k \left[\frac{\bar{m}\delta m}{\omega_k} (c_k^\dagger c_k + d_k^\dagger d_k) + \frac{k\delta m}{R\omega_k} (c_{-k}^\dagger d_k^\dagger + d_k c_{-k}) \right], \text{ where } \omega_k = \sqrt{\bar{m}^2 + \left(\frac{k}{R}\right)^2}. \quad (147)
\end{aligned}$$

Power counting shows that

$$\mathcal{L} \propto \int d^2x [\mathbb{1}\delta m^2 \ln(E_{max}) + o(1/E_{max})] \quad (148)$$

Hence, the expected convergence is

$$E(E_{max}) = a + \frac{b}{E_{max}} \quad (149)$$

in the difference of energies.

In figure 4 the implementation of the free Hamiltonian and the fermion mass term as $H = H_0 + V_{\delta m}$, where the fermion mass is split as $\bar{m} = \delta m = 0.5m$, is shown. The difference of the limits are of order $E/E_0 = o(0.0001)$. In conclusion, the model with extra boson mass converge faster, where the error

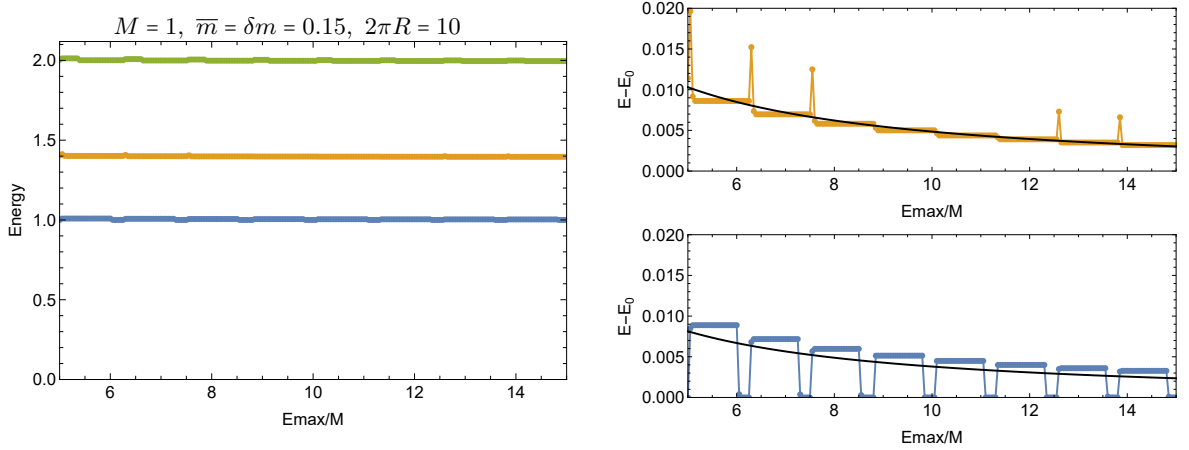


Figure 4: Analogous to the boson diagrams figure 3, the fermion mass is split. The parameter are $R = 10/2\pi$, $M = 1$ and $\bar{m} = \delta m = 0.5m = 0.15$. On the l.h.s. is the difference to the vacuum eigenvalues and the r.h.s. shows additionally the difference to the expected values of the first two eigenvalues. The fit function is $a + b/x$, where a is the convergence limit. It shows, that the 1st level converge to $a_1 = -0.0005 \pm 0.0005$ below the expected value of $E_0 = 1$ with $b_1 \simeq 0.043$, and the 2nd converges to $a = -0.0006 \pm 0.0002$ below $E_0 = 1.39$ with $b_2 \simeq 0.055$. The model converge to the expected values.

shows a $(1/E_{max})^3$ -convergence, while the error due to the fermion extra mass converge as $1/E_{max}$.

¹²It is used that

$$\begin{aligned}
v_{1,k}^* v_{2,k} &= i\sqrt{\omega_k + \frac{k}{R}} \sqrt{\omega_k - \frac{k}{R}} = i\bar{m}_\psi = -v_{2,k}^* v_{1,k} \\
v_{1,-k}^* v_{2,k}^* - v_{2,-k}^* v_{1,k}^* &= -i \left(\sqrt{\omega_k - \frac{k}{R}}^2 - \sqrt{\omega_k + \frac{k}{R}}^2 \right) = 2i\frac{k}{R} = v_{1,k} v_{2,-k} - v_{2,k} v_{1,-k}.
\end{aligned}$$

Either way, having an extra mass term, which is treated as interaction, is a physical freedom. This will be used for boson renormalization

On the other side, the measure points, are not smoothly converging for increasing truncation. Instead the eigenvalues have large energy steps, although the Fock space is increased by just a few states. The fluctuations are decreasing and didn't spoil the convergence. Nevertheless, in figure 4 the eigenvalues are calculated up to a Fock space of 780.814 states, and the fluctuations are still evident.

This shows up, already in the modified free theory, but was not detected in the scalar theory, discussed in[1]. Therefore, it might be due to the appearance of fermions.

A further discussion can be found in section 12.

11.4 Boson Renormalization

11.4.1 Boson Counter Term

The divergent boson loop can be directly evaluated using the Feynman rules eq.(128). The relevant part of $\int d\theta g^2 \sum_{\alpha} \langle f | \phi \bar{\psi} \psi | \alpha \rangle \langle \alpha | \phi \bar{\psi} \psi | i \rangle / (E_i - E_{\alpha})$ is

$$\begin{aligned} \text{Diagram 1: } & \text{A circle with two external dashed lines. Top line labeled } k+n/2, \text{ bottom line labeled } -(k-n/2). \\ & = \frac{-g^2}{2\pi R} \sum_k \frac{\frac{1}{2} - M_{k-n/2, k+k-n/2}}{2\Omega_n(-\Omega_n + \omega_{k-n/2} + \omega_{k+n/2})} \langle f | a_n^\dagger a_n | i \rangle \\ & = \frac{-g^2}{2\pi R} \sum_k \frac{\omega_{k-n/2} \omega_{k+n/2} - m^2 + (k^2 - n^2/4)/R^2}{4\Omega_n \omega_{k-n/2} \omega_{k+n/2} (-\Omega_n + \omega_{k-n/2} + \omega_{k+n/2})} \langle f | a_n^\dagger a_n | i \rangle \end{aligned} \quad (150)$$

$$\begin{aligned} + \text{Diagram 2: } & \text{A circle with two external dashed lines. Top line labeled } n, \text{ bottom line labeled } -(k+n/2). \\ & = \frac{-g^2}{2\pi R} \sum_k \frac{\frac{1}{2} - M_{k-n/2, k+k-n/2}}{2\Omega_n(\Omega_n + \omega_{k-n/2} + \omega_{k+n/2})} \langle f | a_n^\dagger a_n | i \rangle \\ & = \frac{-g^2}{2\pi R} \sum_k \frac{\omega_{k-n/2} \omega_{k+n/2} - m^2 + (k^2 - n^2/4)/R^2}{4\Omega_n \omega_{k-n/2} \omega_{k+n/2} (\Omega_n + \omega_{k-n/2} + \omega_{k+n/2})} \langle f | a_n^\dagger a_n | i \rangle \end{aligned} \quad (151)$$

$$\begin{aligned} + \frac{1}{2} \left[\text{Diagram 3: } & \text{A circle with two external dashed lines. Top line labeled } n, \text{ bottom line labeled } -(k-n/2). \\ & \text{Diagram 4: } \text{A circle with two external dashed lines. Top line labeled } n, \text{ bottom line labeled } -(k+n/2). \right] \\ & = \frac{-g^2}{2\pi R} \sum_k \frac{\frac{1}{2} - M_{k-n/2, k+k-n/2}}{2\Omega_n(\Omega_n + \omega_{k-n/2} + \omega_{k+n/2})} \langle f | a_n^\dagger a_{-n}^\dagger | i \rangle \\ & = \frac{-g^2}{2\pi R} \sum_k \frac{\omega_{k-n/2} \omega_{k+n/2} - m^2 + (k^2 - n^2/4)/R^2}{4\Omega_n \omega_{k-n/2} \omega_{k+n/2} (\Omega_n + \omega_{k-n/2} + \omega_{k+n/2})} \langle f | a_n^\dagger a_{-n}^\dagger | i \rangle \end{aligned} \quad (152)$$

$$\begin{aligned} + \frac{1}{2} \left[\text{Diagram 5: } & \text{A circle with two external dashed lines. Top line labeled } n, \text{ bottom line labeled } -(k+n/2). \\ & \text{Diagram 6: } \text{A circle with two external dashed lines. Top line labeled } n, \text{ bottom line labeled } -(k-n/2). \right] \\ & = \frac{-g^2}{2\pi R} \sum_k \frac{\frac{1}{2} - M_{k-n/2, k+k-n/2}}{2\Omega_n(-\Omega_n + \omega_{k-n/2} + \omega_{k+n/2})} \langle f | a_{-n} a_n | i \rangle \\ & = \frac{-g^2}{2\pi R} \sum_k \frac{\omega_{k-n/2} \omega_{k+n/2} - m^2 + (k^2 - n^2/4)/R^2}{4\Omega_n \omega_{k-n/2} \omega_{k+n/2} (-\Omega_n + \omega_{k-n/2} + \omega_{k+n/2})} \langle f | a_{-n} a_n | i \rangle. \end{aligned} \quad (153)$$

The fermions are contracted in the same way for all diagrams as

$$\leftrightarrow \langle f | \overline{\phi \bar{\psi} \psi \phi \bar{\psi} \psi} | f \rangle \rightarrow (-1)^0,$$

which give no additional sign. The factor of $1/2$ in the last diagrams comes, because the external legs are distinguished in the diagrams, but are actually not distinguishable. Or the $1/2$ can be dropped if one sets $n > 0$ for the last diagrams.

The goal is to identify these diagrams with $V_{\delta M}$ of eq.(144), which is a local operator. In the present form, it is not possible, because the summation of these diagrams is neither Hermitian nor local.

But it is sufficient to choose the counter term, such that the large scale divergence is treated. Neglecting the small scale quantities approximates the diagrams eq.(150) to eq.(153) as

$$A = \frac{1}{2\Omega_n} \cdot \frac{-g^2}{2\pi R} \sum_k \frac{1}{\omega_k} \left(a_n^\dagger a_n + \frac{1}{2} a_n^\dagger a_{-n}^\dagger + \frac{1}{2} a_{-n} a_n \right). \quad (154)$$

By comparing this term with the boson mass interaction eq.(144), one can approximate the contribution to the physical mass as

$$M_{phys}^2 = M^2 - \frac{g^2}{2\pi R} \sum_k \frac{1}{\omega_k} + \text{finite} \quad (155)$$

The mass can be altered by introducing an ancillary mass term, in the form of the mass interaction eq.(144). Keeping $M = \bar{M}$ the bare mass and choosing an additional mass δM to counteract the large scale divergence yields

$$\delta M^2 = \frac{g^2}{2\pi R} \sum_k \frac{1}{\omega_k} \quad (156)$$

$$\Leftrightarrow V_{\delta M} = \frac{g^2}{2\Omega_k \cdot 2\pi R} \sum_k \frac{1}{\omega_k} \left(a_n^\dagger a_n + \frac{1}{2} a_n^\dagger a_{-n}^\dagger + \frac{1}{2} a_{-n} a_n \right). \quad (157)$$

11.4.2 Boson Effective Hamiltonian

Similar to the vacuum, also the corresponding effective Hamiltonian must be calculated in order to add the counter term.

$H_{2,M} \subset H_{eff}$ is added to enlarge the diagrams eq.(150)-eq.(153). Apart from the Heaviside function for the intermediate state, the effective Hamiltonian is identical to these diagrams, yielding

$$\langle f | H_{2,M} | i \rangle = \frac{-g^2}{4\pi R \Omega_n} \sum_k \left[\begin{aligned} & \langle f | a_n^\dagger a_n | i \rangle \frac{\frac{1}{2} - M_{k-n/2, k+k-n/2}}{-\Omega_n + \omega_{k-n/2} + \omega_{k+n/2}} \overset{\triangleright}{\theta}(\omega_{k-n/2} + \omega_{k+n/2} + (E_i - \Omega_n) - E_{max}) \\ & + \langle f | a_n^\dagger a_n | i \rangle \frac{\frac{1}{2} - M_{k-n/2, k+k-n/2}}{\Omega_n + \omega_{k-n/2} + \omega_{k+n/2}} \overset{\triangleright}{\theta}(\omega_{k-n/2} + \omega_{k+n/2} + \Omega_n + E_i - E_{max}) \\ & + \langle f | a_n^\dagger a_{-n}^\dagger | i \rangle \frac{\frac{1}{2} - M_{k-n/2, k+k-n/2}}{\Omega_n + \omega_{k-n/2} + \omega_{k+n/2}} \overset{\triangleright}{\theta}(\omega_{k-n/2} + \omega_{k+n/2} + \Omega_n + E_i - E_{max}) \\ & + \langle f | a_{-n} a_n | i \rangle \frac{\frac{1}{2} - M_{k-n/2, k+k-n/2}}{-\Omega_n + \omega_{k-n/2} + \omega_{k+n/2}} \overset{\triangleright}{\theta}(\omega_{k-n/2} + \omega_{k+n/2} + (E_i - \Omega_n) - E_{max}) \end{aligned} \right] \quad (158)$$

The local approximation is taken analogous to δM in eq.(157). The divergence is cancel, by implementing

$$\begin{aligned} H &= H_0 + H_1 + V_{\delta M} + H_{2,M} \\ \Leftrightarrow H &= H_0 + H_1 + \left[\frac{g^2}{4\pi R \Omega_n} \sum_k \frac{1}{\omega_k} - \frac{g^2}{4\pi R \Omega_n} \sum_k \frac{\overset{\triangleright}{\theta}(2\omega_k - E_{max})}{\omega_k} \right] \left(a_n^\dagger a_n + \frac{1}{2} a_n^\dagger a_{-n}^\dagger + \frac{1}{2} a_{-n} a_n \right), \end{aligned} \quad (159)$$

where a cutoff $\Lambda \gg E_{max}$ is chosen.

For parameter $M = 1$, $m = 0.3$, $R = 10/2\pi$ and a coupling $g = 4$, the first energy levels are shown in figure 5.

The eigenvalues show fluctuations of a similar periodic form, as for the modified free fields in figure 3 or 4. The comparison indicates, that the fluctuations vanish, but at a larger scale as shown in figure 5.

Either way, the expected $1/E_{max}$ -behaviour is evident. The eigenvalues converge to $E = 1.229 \pm 0.011$ and $E = 1.618 \pm 0.014$.

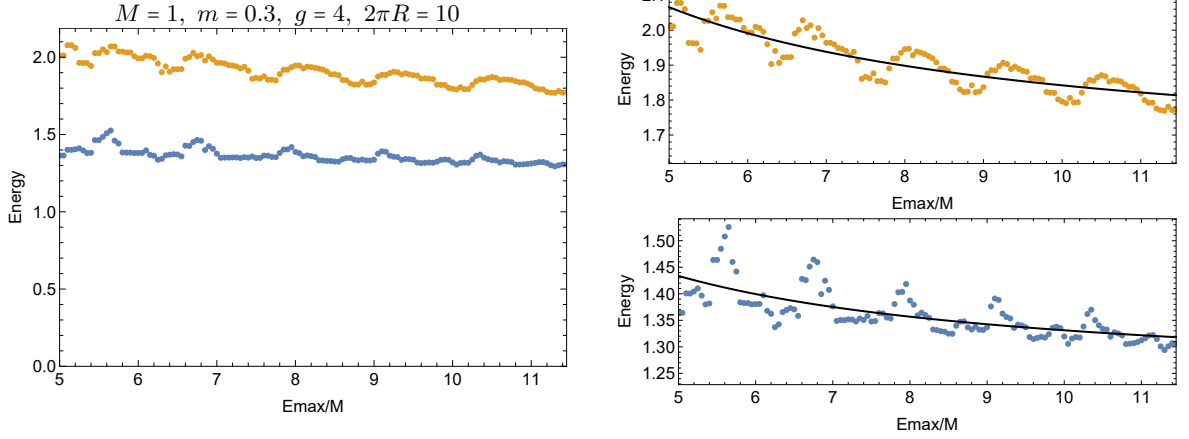


Figure 5: Shown are the first two energy levels of the Hamiltonian, which is corrected by the leading order term of the local approximation. The used parameter are $M = 1$, $m = 0.3$, $R = 10/2\pi$ and $g = 4$ for truncation E_{max} . The renormalization together with $H_2|_{E/E_{max}=0}$ cancel the logarithmic divergence, leaving a $1/E_{max}$ convergence. The r.h.s. shows details of the same data, where a fit of the form $E(E_{max}) = a + b/E_{max}$ is used. The upper fit is $E(E_{max}) = 1.618 + 2.238/E_{max}$ and the lower $E(E_{max}) = 1.229 + 1.024/E_{max}$. Thereby the energy levels converge to $E_1 = 1.229 \pm 0.011$ and $E_2 = 1.618 \pm 0.014$.

11.5 Fermion Effective Hamiltonian

The 2nd order fermion loop diagrams with external fermion legs have a $1/E_{max}$ large scale behaviour (see eq. (129)). $H_{2,m}$ will effectively replace the cutoff, given by truncation, with an independently chosen cutoff Λ , such that the terms converge as $1/\Lambda$, while $\Lambda \rightarrow \infty$

The off diagonal diagrams cancel each other, since

$$\begin{aligned}
 & \left[\text{Diagram 1} \right] = \left[\text{Diagram 2} \right] = \left| \frac{g^2}{2\pi R} \sum_k \frac{\text{sgn}(k+n) \sqrt{\frac{1}{2} - M_{nk}} \sqrt{\frac{1}{2} + M_{nk}}}{2\Omega_{k-n}(\omega_k + \omega_n + \Omega_{k-n})} \right| \\
 & \leftrightarrow \langle 0 | \phi \bar{\psi} \psi \phi \bar{\psi} \psi c^\dagger d^\dagger | 0 \rangle \leftrightarrow \langle 0 | \phi \bar{\psi} \psi \phi \bar{\psi} \psi c^\dagger d^\dagger | 0 \rangle \\
 & \rightarrow (-1)^0 \quad \quad \quad \rightarrow (-1)^3
 \end{aligned}$$

Analogous to the previous cases, the remaining terms are calculated as

$$\begin{aligned}
 & \left[\text{Diagram 3} \right] = \frac{-g^2}{2\pi R} \sum_k \frac{\left(\frac{1}{2} + M_{kn} \right) \hat{\theta}(\Omega_{k-n} + \omega_k + (E_i - \omega_n) - E_{max})}{2\Omega_{k-n}(\Omega_{k-n} + \omega_k - \omega_n)} \langle f | c_n^\dagger c_n + d_n^\dagger d_n | i \rangle \\
 & = \frac{-g^2}{2\pi R} \sum_k \frac{(\omega_k \omega_n + m^2 - \frac{kn}{R^2}) \hat{\theta}(\Omega_{k-n} + \omega_k + (E_i - \omega_n) - E_{max})}{4\Omega_{k-n} \omega_k \omega_n (\Omega_{k-n} + \omega_k - \omega_n)} \langle f | c_n^\dagger c_n + d_n^\dagger d_n | i \rangle \\
 & \leftrightarrow \langle 0 | d \phi \bar{\psi} \psi \phi \bar{\psi} \psi d^\dagger | 0 \rangle \rightarrow (-1)^2
 \end{aligned} \tag{160}$$

$$\begin{aligned}
 & \left[\text{Diagram 4} \right] = \frac{g^2}{2\pi R} \sum_k \frac{\left(\frac{1}{2} - M_{nk} \right) \hat{\theta}(\Omega_{k-n} + \omega_k + \omega_n + E_i - E_{max})}{2\Omega_{k-n}(\Omega_{k-n} + \omega_k + \omega_n)} \langle f | c_n^\dagger c_n + d_n^\dagger d_n | i \rangle \\
 & = \frac{g^2}{2\pi R} \sum_k \frac{(\omega_k \omega_n - m^2 + \frac{kn}{R^2}) \hat{\theta}(\Omega_{k-n} + \omega_k + \omega_n + E_i - E_{max})}{4\Omega_{k-n} \omega_k \omega_n (\Omega_{k-n} + \omega_k + \omega_n)} \langle f | c_n^\dagger c_n + d_n^\dagger d_n | i \rangle \\
 & \leftrightarrow \langle 0 | d \phi \bar{\psi} \psi \phi \bar{\psi} \psi d^\dagger | 0 \rangle \rightarrow (-1)^1
 \end{aligned} \tag{161}$$

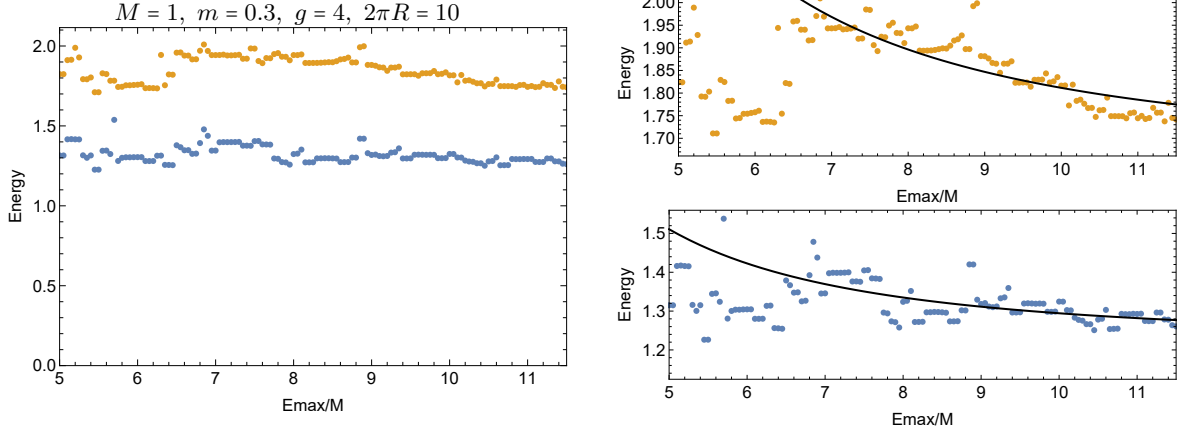


Figure 6: For the shown plots the original Hamiltonian, the second order effective Hamiltonian in the exact formulation, the vacuum and boson renormalization are used. For parameter $M = 1$, $m = 0.3$, $R = 10/2\pi$ and $g = 4$ the energy levels are calculated. The energy levels are expected to converge as $1/E_{max}^2$ beginning at a certain E_{max} value. The data are fitted for $E_{max} > 6.5$ yielding $E(E_{max}) = 1.661 + 15.1/E_{max}^2$ for the upper plot and $E(E_{max}) = 1.122 + 7.87/E_{max}^2$ for the lower. Thereby, the levels converge to $E_1 = 1.222 \pm 0.011$ and $E_2 = 1.661 \pm 0.012$, but the expected convergence behaviour can not be identified cleanly.

The diagrams for particles and antiparticles give the same contribution, which is expected, due to the charge conjugation symmetry of the Yukawa interaction.

Adding together all terms of H_2 in the exact form and the counter terms (eq.(142), eq.(158), eq.(159), eq.(160), eq.(161)), yields the Hamiltonian

$$\begin{aligned}
 H &= H_0 + H_1 + (H_{2,V} + \delta_V) + H_{2,M} + V_{\delta M} + H_{2,m}, \quad \text{with} \quad (162) \\
 \langle f | H_{2,V} + \delta_V | i \rangle &= \frac{g^2}{2\pi R} \sum_{n,k} \frac{\left(\frac{1}{2} - M_{k-n/2, k+k-n/2}\right) \overset{\leftarrow}{\theta}(E_i + \Omega_n + \omega_{k-n/2} + \omega_{k+n/2} - E_{max})}{2\Omega_n(\Omega_n + \omega_{k-n/2} + \omega_{k+n/2})} \langle f | i \rangle \\
 \langle f | V_{\delta M} | i \rangle &= \sum_n \frac{g^2}{4\pi R \Omega_n} \sum_k \frac{1}{\omega_k} \langle f | a_n^\dagger a_n + \frac{1}{2} a_n^\dagger a_{-n}^\dagger + \frac{1}{2} a_{-n} a_n | i \rangle \\
 \langle f | H_{2,M} | i \rangle &= \sum_n \frac{-g^2}{4\pi R \Omega_n} \sum_k \left[\frac{\left(\frac{1}{2} - M_{k-n/2, k+k-n/2}\right) \overset{\rightarrow}{\theta}(\omega_{k-n/2} + \omega_{k+n/2} + (E_i - \Omega_n) - E_{max})}{-\Omega_n + \omega_{k-n/2} + \omega_{k+n/2}} \langle f | a_n^\dagger a_n | i \rangle \right. \\
 &\quad + \frac{\left(\frac{1}{2} - M_{k-n/2, k+k-n/2}\right) \overset{\rightarrow}{\theta}(\omega_{k-n/2} + \omega_{k+n/2} + \Omega_n + E_i - E_{max})}{\Omega_n + \omega_{k-n/2} + \omega_{k+n/2}} \langle f | a_n^\dagger a_n | i \rangle \\
 &\quad + \frac{\left(\frac{1}{2} - M_{k-n/2, k+k-n/2}\right) \overset{\rightarrow}{\theta}(\omega_{k-n/2} + \omega_{k+n/2} + \Omega_n + E_i - E_{max})}{\Omega_n + \omega_{k-n/2} + \omega_{k+n/2}} \langle f | a_n^\dagger a_{-n}^\dagger | i \rangle \\
 &\quad \left. + \frac{\left(\frac{1}{2} - M_{k-n/2, k+k-n/2}\right) \overset{\rightarrow}{\theta}(\omega_{k-n/2} + \omega_{k+n/2} + (E_i - \Omega_n) - E_{max})}{-\Omega_n + \omega_{k-n/2} + \omega_{k+n/2}} \langle f | a_{-n} a_n | i \rangle \right] \\
 \langle f | H_{2,m} | i \rangle &= \sum_n \frac{g^2}{2\pi R} \sum_k \left[-\frac{\left(\frac{1}{2} + M_{kn}\right) \overset{\rightarrow}{\theta}(\Omega_{k-n} + \omega_k + (E_i - \omega_n) - E_{max})}{2\Omega_{k-n}(\Omega_{k-n} + \omega_k - \omega_n)} \langle f | c_n^\dagger c_n + d_n^\dagger d_n | i \rangle \right. \\
 &\quad \left. + \frac{\left(\frac{1}{2} - M_{nk}\right) \overset{\rightarrow}{\theta}(\Omega_{k-n} + \omega_k + \omega_n + E_i - E_{max})}{2\Omega_{k-n}(\Omega_{k-n} + \omega_k + \omega_n)} \langle f | c_n^\dagger c_n + d_n^\dagger d_n | i \rangle \right]
 \end{aligned}$$

The implementation is shown in figure 6 for the parameter $M = 1$, $m = 0.3$, $R = 10/2\pi$ and $g = 4$.

It is difficult to identify the expected $1/E_{max}^2$ convergence. Therefore, the point where the data start to converge can not be identified cleanly. Here $E_{max} > 6.5$ is chosen to be matched with the fit function

$$E(E_{max}) = a + b \left(\frac{1}{E_{max}} \right)^2.$$

The energy levels converge to $a_1 = 1.123 \pm 0.022$ and $a_2 = 1.661 \pm 0.012$. One can compare this with the data from the local approximation shown in figure 5, where $a_1^{(0)} = 1.229 \pm 0.011$ and $a_2^{(0)} = 1.618 \pm 0.014$. The results are close, but the error ranges are not overlapping.

Note that, in the matching with the function, the starting point of the fit is chosen by hand. Especially because the $1/E_{max}^2$ convergence is difficult to identify, this freedom can introduce a bias in the results. That is why the calculations in the local approximation seem more suitable for the discussed problem.

The small deviation of both methods might be due to that problem, but in principal the local approximation $H_2|_{E/E_{max}=0}$ with a $1/E_{max}$ -convergence or the exact description H_2 with a $(1/E_{max})^2$ -convergence lead to the same results.

A detailed comparison of both methods is done in the next chapter.

12 Exact Calculations Versus Local Approximation

The motivation of dropping the local approximation is, that in the description of [2], it was unclear, whether the power counting arguments are valid, and whether the exact Hamiltonian can be implemented as in eq.(162).

But the power counting holds as shown in chapter 11.1 and the implementation is efficient as outlined in the next section.

Despite that, additional difficulties appear in the exact formulation, which is discussed later in this chapter.

12.1 Implementation

The problem of the implementation is that the Heaviside function needs to be evaluated for each state. For a large basis one has to find a efficient way to do the calculations.

The concept is to insert the Heaviside as a matrix instead of a function. In the examples of H_2 , the intermediate state depends on two kinetic variables $E_\alpha = E_\alpha(k, n)$. The definition of H_2 (see eq.(136)) indicates, that the theta function is in the form

$$\vec{\theta}(E_\alpha(n, k) + E_i - E_{max}) \equiv \vec{\theta}_{nki}(E_{max}). \quad (163)$$

The function depends only on discretized variables and can be written as a matrix, which is generated before the elements H_{ji} are calculated. Note that, one could take $\vec{\theta}_{nk}$ as a vector (i is the vector index here). This way, theta can act on all states at once instead of being evaluated individually.

In the end, the generation of the whole Hamiltonian H_{eff} is still negligible compared to the calculations of eigenvalues.

12.2 Additional Difficulties

Nevertheless, the result of the exact effective Hamiltonian, shown in figure 6, is less suitable to deduce the convergence compared to the leading order term of the local approximation shown in 5. This matter will be discussed and other disadvantages are pointed out.

The problem arises from the multiple particle Feynman rules as described in section 10. For an exact calculation, multiplicity for intermediate states must be taken into account. In appendix 18 it is derived, that the multiple particle rules reduce to the single particle rules, if all diagrams are considered.

Therefore, the calculations for H_2 in section 11 are only valid, if H_2 is included completely. Because of the multiplicity for intermediate states, the different parts of H_2 should not be viewed as separated.

On the other side, for the expansion around energies $E \ll E_{max}$, the single particle Feynman rules are sufficient. Thus, all terms of H_2 can be included separately. In particular one could just include the boson renormalization and ignore the vacuum and fermion correction. As shown in figure 5 this already give the expected, first order convergence.

For the exact calculation, including only the boson correction, or for example ignoring the fermion correction don't yield to the expected convergence. In both cases the power counting arguments would give a $1/E_{max}$ convergence, if the diagrams would be separate. The results are shown in figure 7.

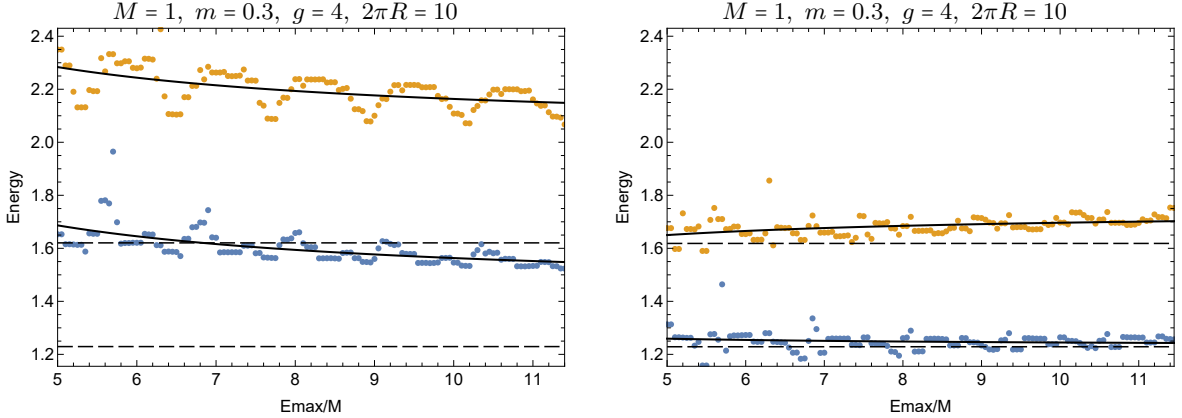


Figure 7: On the l.h.s. the energy levels for the Hamiltonian with the exact boson effective Hamiltonian at 2nd order and the boson counter term are calculated, i.e. $H = H_0 + H_1 + H_{2,M} + V_{\delta M}$. On the r.h.s. also the exact calculations for the vacuum, i.e. $H_{2,V} + \delta_V$, are added to the Hamiltonian. The dashed lines show the expected value of the convergence, coming from the leading order local approximation as shown in figure 5. Moreover, the fit function $a + b/E_{max}$ is used. On the l.h.s. the energies converge to $a_1 = 1.440 \pm 0.018$ with $b_1 \simeq 1.2$ and $a_2 = 2.043 \pm 0.023$ with $b_2 \simeq 1.2$, which is not in agreement with $a_1^{(0)} = 1.229 \pm 0.011$ and $a_2^{(0)} = 1.618 \pm 0.014$ from the local approximation. Adding the vacuum narrow the difference as $a_1 = 1.230 \pm 0.012$ with $b_1 \simeq 0.14$ and $a_2 = 1.742 \pm 0.012$ with $b_2 \simeq -0.46$, but especially for the 2nd excited state, the disagreement is evident. The used parameter are $M = 1$, $m = 0.3$, $R = 10/2\pi$ and $g = 4$.

The energy level don't converge to the same values as the local approximation, which are $E_1 = 1.229 \pm 0.011$ and $E_2 = 1.618 \pm 0.014$.

Only the boson correction, i.e. $H = H_0 + H_1 + H_{2,M} + V_{\delta M}$, yields $E_1^{bos} = 1.440 \pm 0.018$ and $E_2^{bos} = 2.043 \pm 0.023$. Including the vacuum part reduce the deviation to $E_1^{bos+vac} = 1.230 \pm 0.012$ and $E_2^{bos+vac} = 1.742 \pm 0.012$.

But only, if all loop diagrams of H_2 are included, both methods give comparable results as shown in figure 6 of the previous chapter.

The expected $(1/E_{max})^2$ convergence could not be cleanly identified, even when all loop diagrams are included. This could be due to the small scale fluctuations of the data points, which obscure the large scale behaviour.

From one truncation energy E_{max} to the next $E_{max} + \epsilon$ the difference are a few states at the boundary. A fluctuation means, that the eigenvalues change by a non continuous amount, even when ϵ is small. That means

$$E(E_{max} + \epsilon) - E(E_{max}) \gg \epsilon.$$

Therefore, the fluctuations are due to boundary effects.

Note that, in the chapter 11.3, where the masses were split and treated as interaction, the fluctuations already appear. The results indicate, that the fluctuations are expected to decrease for large E_{max} , but that happens at a larger energy scale as shown in the plots.

In an exact calculation these effects would be treated by the part of the effective Hamiltonian corresponding to

$$H_2 \supset \int d\theta (g^2 \phi \bar{\psi} \psi + g^2 \bar{\psi} \psi \bar{\psi} \psi), \quad (164)$$

which are tree level diagrams.

In conclusion, the theory do have visible boundary effects and the multiple particle diagrammatic rules are only valid, if all diagrams are considered. Therefore, the exact calculations are less usable as long as not all parts consistent with an expansion in the coupling are included. Also boundary terms must be taken into account.

From eq.(164) it follows, that there are 32 terms, which would be needed additionally.

On the other side, for an approximation around $E/E_{max} = 0$ boundary effects can be ignored. That is why the leading order local approximation is more convenient for the subject discussed here and will be used for the next chapter.

But, if one is interested in the limit of scalar mass $M \rightarrow \infty$, which yields the convergence to the Gross Neveu model. Then, the Yukawa interaction couples states with others, outside the low energy regime $E \lesssim E_{max}$, such that the local approximation might be less precise. For future work it could be necessary to pursue with the introduced method.

For this, the next step is to complete H_2 with the missing boundary terms. Then, the convergence for varying M must be compared with the local approximation. For small M the results should be recovered, but larger M might give a better convergence. Probably, a smaller Fock space is needed for the exact calculation.

The limits for $E_{max} \rightarrow \infty$ for different M need to be determined by a fit. In the end, these values can be displayed in a diagram, which gives the energy spectrum for varying M .

13 Basis for Varying Energies and Volumes

This chapter gives a quick overview of the number of states and the time needed for different parameter.

Truncation has the property, that the number of states scale exponentially[1]. This is shown in figure 8. The method for the implementation is outlined in appendix 17. Since the program is sufficient fast,

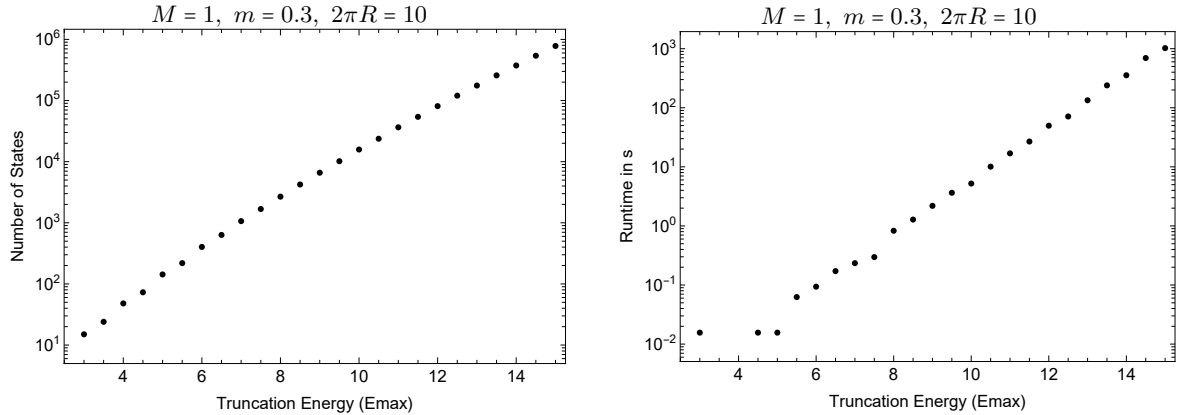


Figure 8: The Fock basis is generated for different truncation energies. All states have momentum zero, fermion charge zero and are charge even. The number of states and the required time to generate the basis scales exponentially.

the restriction of the available memory space turned out to be more important than the runtime. The computations were performed by a basic setup with 20GB of RAM.

The bottle neck is the calculation of eigenvalues. In the plots E_{max} was increased until the program shuts down. This gives an effective limit of around 50 000 states for the Yukawa interaction. This corresponds to the parameter $M = 1$, $m = 0.3$, $2\pi R = 10$ and $E_{max} = 11.5$.

For future work, one might be interested in increasing the volume instead. The single particle energy reads

$$E = \sqrt{m^2 + \left(\frac{n}{R}\right)^2} \rightarrow E \cdot R \simeq n,$$

where only large scale quantities are considered. Therefore, increasing R is expected to scale in the same way as E_{max} . This is shown in figure 9. The exponential growth in the number of states is

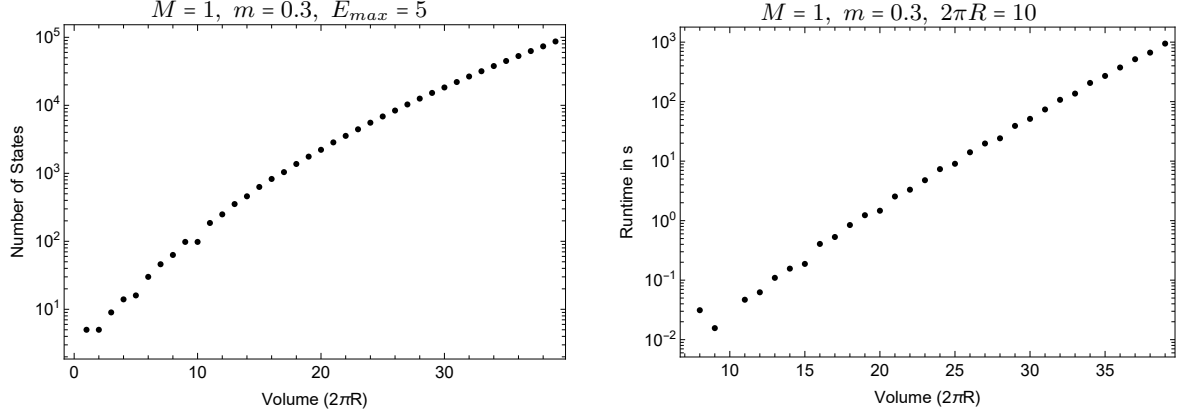


Figure 9: Shown is the scaling of the Fock space for increasing volume. The states are restricted by the quantum numbers of momentum $L = 0$, fermion charge $Q = 0$ and charge conjugation $C = +1$. The number of states and the timing grow exponentially, although the number have a slight deviation from the expectation.

flattening a bit for larger R , but the timing increases as expected. Note that the maximal volume calculated is $2\pi R = 39$ for $E_{max} = 5$, $M = 1$, $m = 0.3$ with around 87 000 states. For $2\pi R = 40$ the program run out of memory space and shuts down. Therefore, it is difficult to calculate larger volumes for the current implementation.

14 Discussion

In this section the spectrum for varying coupling is analysed. The results are compared with the paper of Brooks and Frautschi[3]. For comparison the bare Hamiltonian $H = H_0 + H_1$ without corrections is used.

Figure 10 shows on the l.h.s. the energy spectrum for varying coupling up to $g = 4$ and fixed truncation $E_{max} = 13$. On the r.h.s. are the results from [3].

In both plots the same parameter of $M = 1$, $m = 0.3$ and $R = 1$ are used.

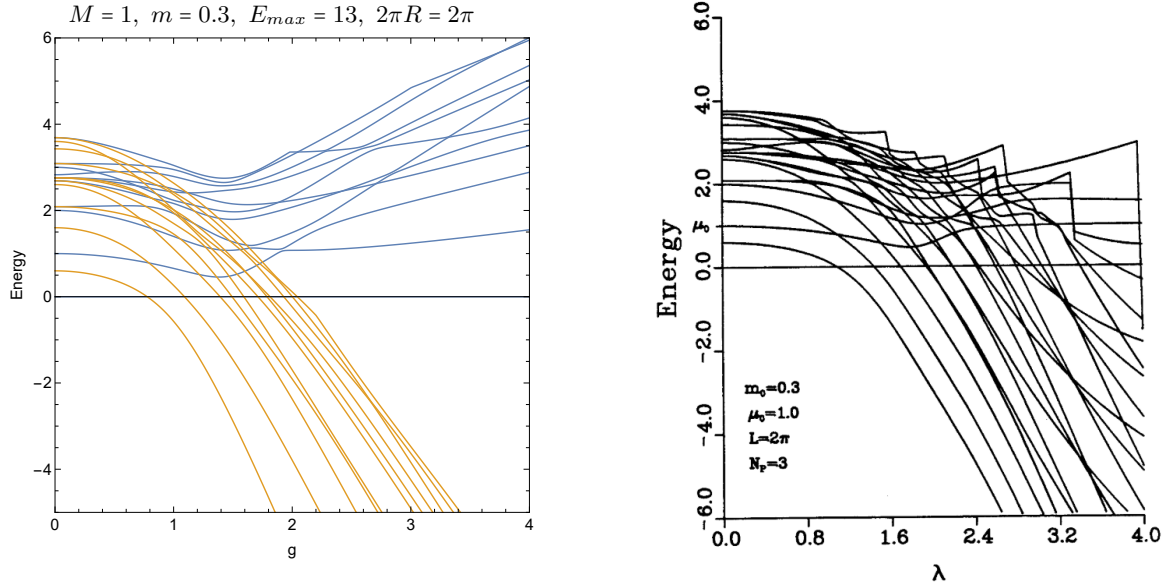


Figure 10: On the l.h.s. the eigenvalues for varying coupling and $E_{max} = 13$, $M = 1$, $m = 0.3$, $R = 1$ are calculated. The C-even and C-odd sectors are computed separately, where the C-even states are blue marked and the C-odd orange. The vacuum, defined as the lowest C-even state, is subtracted. This is to be compared to the results of Brooks and Frautschi [3] (r.h.s.), where a similar truncation method is used. The difference is, that a basis state can only contain $N_F = 3$ bosons of the same momentum. The results of [3] are not regained, where the deviation grows with larger couplings.

The vacuum is defined as the lowest energy charge conjugate even state. The other eigenvalues are subtracted by this state.

In both computations the energy of at least one state becomes negative. Using, that the described implementation distinguishes between c-odd and c-even states, the negative energies can be identified with c-odd states.

If one would define the vacuum as the lowest energy state, the crossing of the vacuum level would indicate a phase transition. But the interaction between the two sectors are forbidden by charge conjugation symmetry and are viewed as separate. If an interaction between those states would be added, the phase transition might vanish.

The two results don't match, where the disagreement grows with larger coupling. For example one can compare the eigenvalues for the state with free energy $E(g=0) = 1$, which is associated with the boson.

Since the eigenvalues change continuously, the point around $g = 2$ is a crossing point with another state. Then, the energy of the boson level increase for higher coupling.

On the other hand the diagram of Brooks and Frautschi indicates, that the energy converges to $E = 1$. Thus, the energy levels of the present implementation are increased for larger coupling.

The disagreement might be caused by an approximation used in [3]. The truncation cutoff as described in previous chapters is used in the same way. But due to computational limitations, Brooks and Frautschi used a restriction on the maximal number of boson excitations. In this case $N_P = 3$ indicates that a basis state can not contain more than three bosons of the same momentum.

One can conclude, that the approximation is less accurate for higher coupling. Hence the implementation presented, can be seen as an enhancement of the approximation.

Moreover, there are another difference at large coupling. Both plots have energy levels with a discontinuous behaviour. The reason is that only a certain amount of the lowest eigenvalues are calculated. Thereby, the changes indicate a crossing with higher energy states.

In this case, the version of the described implementation shows 11 C-even and 11 C-odd states, while the version of Brooks and Frantschi shows mostly C-odd states for large coupling. This makes the plot look very different for $g \rightarrow 4$.

To be consistent, the boson needs to be renormalized. This is done using the computations for leading order local approximation, i.e.

$$H = H_0 + H_1 + V_{\delta M} + H_2 \Big|_{E/E_{max}=0}.$$

The eigenvalues for varying coupling are shown in figure 11.

Due to the corrections, all energy levels are lifted with respect to the previous result. The difference grows for larger coupling.

For future work, it would be interesting to modify the implementation, such that Majorana fermions are described. By doing so, the method can be tested against the paper of Anand et al.[5].

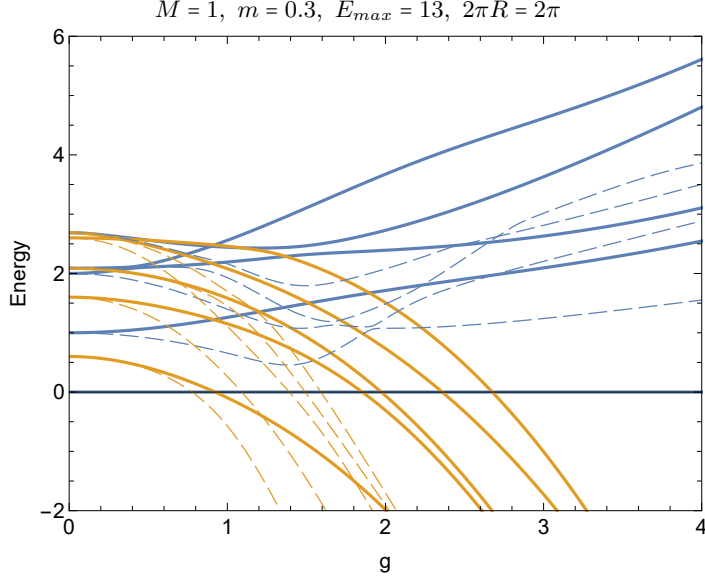


Figure 11: Shown are the energy spectrum for varying couplings. The local approximation at leading order is used, i.e. $H = H_0 + H_1 + V_{\delta M} + H_2|_{E/E_{max}=0}$. The C-even states are marked blue and the C-odd states orange. The lowest C-even state is defined to be the vacuum and is subtracted. Additionally the thin, dashed lines show the spectrum of the not renormalized theory as shown in figure 10.

15 Conclusion

The systematics of quantizing and implementing the Yukawa model with Hamiltonian truncation were explained. It was shown, how symmetries can be used to reduce the computations. In particular charge conjugation symmetry was utilized and the implementation of doublets was explained.

The energy spectrum of the unmodified Hamiltonian did not converge. Thereby, the UV divergent property of the Yukawa coupling was shown. Furthermore, by distinguishing C-even and C-odd states, it was shown, that C-odd states can have lower energy than the C-even vacuum (i.e. the normal vacuum for vanishing coupling), but the transition is forbidden by charge conjugation symmetry. Thus, C-even and C-odd states can be viewed as independent.

By using the concept of Delouche et al.[2], the theory was renormalized. In doing so, the effective Hamiltonian in the description of Cohen et al.[1] was implemented. The leading order of the local expansion, i.e. $E/E_{max} = 0$ – taking all small scale quantities to zero, was taken. As expected from power counting arguments, this resulted in a $1/E_{max}$ convergence of the energy spectrum. Although, the coarse behaviour was expected, periodic fluctuations of the eigenvalues appeared in the calculations. This was compared to a description, equivalent to the free theory. In there, the masses were split to get additional interaction terms. Already in this theory the fluctuations were evident. In comparison with this model, the fluctuations are expected to vanish slowly.

In the diagrams for varying coupling, the results of Brooks and Frautschi[3] were not regained. The increasing error for larger coupling, are due to an approximation used in their paper.

Additionally, a modification of the effective Hamiltonian was explored. The local approximation was dropped, leaving an expansion in the coupling only. Although the power counting arguments are still valid and the implementation is efficient, the new method turned out to be less suitable to calculate the low energy spectrum. For the new method multi-particle Feynman rules were derived. This led to the conclusion, that boundary terms of the effective Hamiltonian must be taken into account. This enlarge the computations with respect to the local approximation, since these terms can be ignored in there. Nevertheless, both methods yielded similar results, but the exact formulation was less precise without the boundary terms.

In conclusion, the first step in implementing the Gross-Neveu model using Yukawa interactions in truncation is done.

16 Literature

- [1] T.Cohen, K.Farnsworth, R.Houtz, M.A.Luty: „*Hamiltonian Truncation Effective Theory*“, arXiv:2110.08273
- [2] O.Delouche, J.E.Miro, J.Ingolby: „*Hamiltonian Truncation Crafted for UV-divergent QFTs*“, arXiv:2312.09221v2
- [3] E.D.Brooks III, S.C.Frautschi: „*Scalars Coupled to Fermions in 1+1 Dimensions*“, (California Institute of Technology, Pasadena), Z.Phys.C-Particles and Fields **23**, 263-273 (1984)
- [4] D.J.Gross, A.Neveu: „*Dynamical symmetry breaking in asymptotically free field theories*“, (Institute for Advanced Study, Princeton, New Jersey), Physical Review D **10** (1974), 085440
- [5] N.Anand, A.L.Fitzpatrick, E.Katz, Z.U.Khandker, M.T.Walters, Y.Xin: „*Introduction to Light-cone Conformal Truncation: QFT Dynamics from CFT Data*“, arXiv:2005.13544
- [6] S.Coleman: „*Aspects of Symmetry*“, Cambridge University Press, New York, USA, (1985)
- [7] .Kodad: „*The Magic of Hashing for Efficient Data Retrieval*“, Medium, https://medium.com/@serene_mulberry_tiger_125/the-magic-of-hashing-for-efficient-data-retrieval-37bed5c927a5 (3.9.2024)
- [8] L.Polley and D.E.L.Pottinger: „*Variational Calculations in Quantum Field Theory*“, World Scientific, (1988)
- [9] Robert M.Wald: „*General Relativity*“, University of Chicago Press, Chicago, USA, ISBN: 0-226-87033-2, (1984)
- [10] Steven Weinberg: „*The Quantum Theory of Fields*“, Cambridge University Press, New York, USA, ISBN: 0 521 55001 7, (1995)
- [11] P.Di Francesco, P. Mathieu, D. Séméchal: „*Conformal Field Theory*“, Springer Verlag, New York, USA, ISBN: 13:978-1-4612-7475-9
- [12] T.Lancaster, S.J.Blundell: „*Quantum Field Theory for the Gifted Amateur*“, Oxford University Press, Oxford, United Kingdom, ISBN: 978-0-19-969932-2, (2014)
- [13] M.Maggiore: „*A Modern Introduction to Quantum Field Theory*“, Oxford University Press, Oxford, UK, ISBN: 0 19 852073 5, (2005)
- [14] D.j.Griffiths, D.F.Schroeter: „*Introduction to Quantum Mechanics*“, Cambridge University Press, Cambridge, UK, ISBN: 978-1-107-18963-8, (2018)
- [15] M.D.Schwartz: „*Quantum Field Theory and the Standard Model*“, Cambridge University Press, Cambridge, UK, ISBN: 978-1-107-03473-0, (2014)
- [16] H.Georgi: „*Effective Field Theory*“, Annual Review of Nuclear and Particle Science, **43** (1993)
- [17] J.Elias-Miro, M.Montull, M.Riembau “*The renormalized Hamiltonian truncation method in the large E_T expansion*“, JHEP, **4**, arXiv: 1512.06901, (2016)
- [18] D.Poland,S.Rychkow,A.Vichi: „*The Conformal Bootstrap: Theory, Numerical Techniques, and Applications*“, Rev.Mod.Phys.,**91**, 015002, (2019)
- [19] M.P.Hernández, : „*1 Lattice Field Theory Fundamentals* “, Modern Perspectives in Lattice QCD: Quantum Field Theory and High Performance Computing: Lecture Notes of the Les Houches Summer School, **93**, Oxford University Press, Oxford, UK

17 Appendix A: Basis Generation

The number of basis states are reduced by restricting quantum numbers to certain values. Truncation defines the energy $E \leq E_{max}$. Moreover L the momentum eigenvalue of $\hat{L}$ and the number of fermions minus anti-fermions Q must be zero.

The basis is symbolically

$$\{a^\dagger\} \otimes \{c^\dagger\} \otimes \{d^\dagger\} \Big|_{E \leq E_{max}, L=0, Q=0}. \quad (165)$$

$\{a^\dagger\}$ is short hand for all linear independent states, which can be created using boson operators (similar $\{c^\dagger\}$ and $\{d^\dagger\}$) and $\otimes$ is the outer product of those states. The product will combine complete Fock bases formed from restricted Hilbert spaces, where the different sets don't intersect (except for the vacuum). Thereby the outer product trivially combines all states, forming a complete basis in a enlarged Fock space.

The idea of fasten the generation, is to generate small subsets of the basis and consider the restrictions, when the outer products are taken.

Here the subsets

$$\left\{ \begin{matrix} a^\dagger \\ \hat{L} > 0 \end{matrix} \right\}, \left\{ \begin{matrix} a^\dagger \\ \hat{L} = 0 \end{matrix} \right\}, \left\{ \begin{matrix} c^\dagger \\ \hat{L} > 0 \\ \hat{Q} > 0 \end{matrix} \right\}, \left\{ \begin{matrix} c^\dagger \\ \hat{L} = 0 \\ \hat{Q} > 0 \end{matrix} \right\} \quad (166)$$

are used to create the whole basis. Additional the Hilbert space of each particle in each subset is reduced. E.g. $\hat{L} > 0$ means, that only creation operators of positive momentum are used. The total momentum could still be zero, since every subset contain the vacuum. Analogous $\hat{Q} > 0$ indicates that only $\{c^\dagger\}$ is used.

The subsets itself are not restricted by the quantum numbers of the final basis. Their properties are altered by merging with other subsets. For example, if a state of a subset have overall momentum L and energy E , then the other particles added later must also have momentum L and a energy $E_{kin} + E_{pot}$ smaller then $E_{max} - E$. Then, a state of an arbitrary subset, can only be part of the final basis, if

$$\begin{aligned} E_{kin} + E_{pot} &\leq E_{max} - E \\ \Rightarrow E + L/R &\leq E_{max} \end{aligned} \quad (167)$$

is satisfied. The l.h.s. of the first line refers to the part, which is added to form the final basis state. Then, the first line is used to deduce a restriction on the incomplete state itself.

The computations of the outer product is sensitive to the number of states. That is why this condition is applied whenever new sets are created.

The first step to merge is

$$\left\{ \begin{matrix} a^\dagger \\ \hat{L} > 0 \end{matrix} \right\}, \left\{ \begin{matrix} a^\dagger \\ \hat{L} = 0 \end{matrix} \right\} \rightarrow \{a^\dagger\} \quad \left| \quad \prod_{k,l} \left\{ \begin{matrix} a^\dagger \\ L = k \geq 0 \end{matrix} \right\} \otimes \left\{ \begin{matrix} a^\dagger \\ L = l \geq 0 \end{matrix} \right\} \otimes \left\{ \begin{matrix} a^\dagger \\ L = 0 \end{matrix} \right\} = \left\{ \begin{matrix} a^\dagger \\ L = k - l \end{matrix} \right\} \right. \quad (168)$$

$$\left\{ \begin{matrix} c^\dagger \\ \hat{L} > 0 \\ \hat{Q} > 0 \end{matrix} \right\}, \left\{ \begin{matrix} c^\dagger \\ \hat{L} = 0 \\ \hat{Q} > 0 \end{matrix} \right\} \rightarrow \left\{ \begin{matrix} c^\dagger \\ \hat{Q} > 0 \end{matrix} \right\} \quad \left| \quad \prod_{k,l} \left\{ \begin{matrix} c^\dagger \\ L = k \geq 0 \\ Q = h \geq 0 \end{matrix} \right\} \otimes \left\{ \begin{matrix} c^\dagger \\ L = l \geq 0 \\ Q = i \geq 0 \end{matrix} \right\} \otimes \left\{ \begin{matrix} c^\dagger \\ L = 0 \\ Q = j \geq 0 \end{matrix} \right\} = \left\{ \begin{matrix} c^\dagger \\ L = k - l \\ Q = h + i + j \end{matrix} \right\}. \quad (169)$$

Taking the outer product also includes calculating the new quantum numbers for each state. Thereby the notation on the r.h.s. refer to overall quantum number of individual states, while on the l.h.s. the operators, which create those states are specified. Thus, the l.h.s. restricts the Hilbert spaces and the r.h.s refer to the Fock space.

It is used that the subset with positive momentum is equivalent to the subset of negative momentum. In the merging the momentum of the latter is taken negative.

The fermion part is concluded by merging

$$\left\{ \begin{matrix} c^\dagger \\ \hat{Q} > 0 \end{matrix} \right\} \rightarrow \{c^\dagger\} \otimes \{d^\dagger\}|_{Q=0} \left| \prod_i \left\{ \begin{matrix} c^\dagger \\ L=k \\ Q=i > 0 \end{matrix} \right\} \otimes \left\{ \begin{matrix} c^\dagger \\ L=l \\ Q=i > 0 \end{matrix} \right\} = \left\{ \begin{matrix} c^\dagger, d^\dagger \\ L=k+l \\ Q=i-i=0 \end{matrix} \right\} \right., \quad (170)$$

where Q is taken negative in the second subset, i.e. $c^\dagger \rightarrow d^\dagger$. Note that, taking the outer product here is faster than the previous ones, since the product runs only over one variable.

The same is true for the final merging:

$$\{a^\dagger\}, \{c^\dagger\} \otimes \{d^\dagger\} \rightarrow \{a^\dagger\} \otimes \{c^\dagger\} \otimes \{d^\dagger\}|_{L=0} \left| \prod_k \left\{ \begin{matrix} a^\dagger \\ L=k \end{matrix} \right\} \otimes \left\{ \begin{matrix} c^\dagger, d^\dagger \\ L=k \end{matrix} \right\} = \left\{ \begin{matrix} a^\dagger, c^\dagger, d^\dagger \\ L=k-k=0 \end{matrix} \right\} \right., \quad (171)$$

where the momenta of the second states in the outer product are taken negative.

If condition eq.(167) is applied as usual after each merging, every state in the basis satisfy $E \leq E_{max}$, $L=0$ and $Q=0$.

18 Appendix B: Calculations

18.1 Contracted Vertex Factor

Using the definitions in eq.(49), it is straight forward to calculate the fundamental factors, which appear in eq.(53). The expression of the $c_l^\dagger c_k$ -term is

$$\begin{aligned} \mathbf{v}_l^\dagger \gamma^0 \mathbf{v}_k &= \begin{pmatrix} \sqrt{\omega_l + \frac{l}{R}} & -i\sqrt{\omega_l - \frac{l}{R}} \end{pmatrix} \begin{pmatrix} -i \\ i \end{pmatrix} \begin{pmatrix} \sqrt{\omega_k + \frac{k}{R}} \\ i\sqrt{\omega_k - \frac{k}{R}} \end{pmatrix} \\ &= \sqrt{(\omega_l + \frac{l}{R})(\omega_k - \frac{k}{R})} + \sqrt{(\omega_l - \frac{l}{R})(\omega_k + \frac{k}{R})} \\ &= \left[2\omega_k \omega_l - 2\frac{kl}{R^2} + 2\sqrt{(\omega_k + \frac{k}{R})(\omega_k - \frac{k}{R})(\omega_l + \frac{l}{R})(\omega_l - \frac{l}{R})} \right]^{1/2} \\ &= \left[2\omega_k \omega_l - 2\frac{kl}{R^2} + 2m^2 \right]^{1/2} \\ &\Leftrightarrow \frac{\mathbf{v}_l^\dagger \gamma^0 \mathbf{v}_k}{\sqrt{4\omega_k \omega_l}} = \sqrt{\frac{1}{2} + M_{kl}}, \quad \text{with } M_{kl} = \frac{m^2 - kl/R^2}{2\omega_k \omega_l}. \end{aligned} \quad (172)$$

Using $(\gamma^0)^T = -\gamma^0$ the factor for the $d_k d_l^\dagger$ -term is

$$\mathbf{v}_k^T \gamma^0 \mathbf{v}_l^* = -\mathbf{v}_l^\dagger \gamma^0 \mathbf{v}_k = -\sqrt{4\omega_k \omega_l} \sqrt{\frac{1}{2} + M_{kl}}. \quad (173)$$

The other factor is

$$\mathbf{v}_l^T \gamma^0 \mathbf{v}_{-k} = \sqrt{(\omega_l + \frac{l}{R})(\omega_k + \frac{k}{R})} - \sqrt{(\omega_l - \frac{l}{R})(\omega_k - \frac{k}{R})} \quad (174)$$

$$\Leftrightarrow |\mathbf{v}_l^T \gamma^0 \mathbf{v}_{-k}| = \left[2\omega_k \omega_l + 2\frac{kl}{R^2} - 2m^2 \right] \quad (175)$$

The form is similar to the previous case, but there is an additional sign, which must be calculated separately.

The l.h.s of expression eq.(174) is positive, if the first square root is bigger then the second. In detail:

$$\begin{aligned}
& \sqrt{(\omega_l + \frac{l}{R})(\omega_k + \frac{k}{R})} > \sqrt{(\omega_l - \frac{l}{R})(\omega_k - \frac{k}{R})} \\
& \Leftrightarrow (\omega_l + \frac{l}{R})(\omega_k + \frac{k}{R}) > (\omega_l - \frac{l}{R})(\omega_k - \frac{k}{R}) \\
& \Leftrightarrow l\omega_k > -k\omega_l
\end{aligned} \tag{176}$$

This condition is trivially satisfied, if $k, l > 0$ and falsified if $k, l < 0$, which indicates that the r.h.s is proportional to $\text{sgn}(k+l)$. It turns out, that it is also true for the cases $l > 0, k < 0$ and $l < 0, k > 0$. The equation get solved by squaring both sides, but the second case gets a minus sign:

$$\begin{aligned}
& \pm \left(m^2 + \left(\frac{k}{R} \right)^2 \right) l^2 > \pm \left(m^2 + \left(\frac{l}{R} \right)^2 \right) k^2 \\
& \Leftrightarrow \pm l^2 > \pm k^2 \\
& \Leftrightarrow |l| - |k| = l + k > 0, \text{ for } l > 0, k < 0; \\
& -|l| + |k| = l + k > 0, \text{ for } l < 0, k > 0.
\end{aligned} \tag{177}$$

$$\tag{178}$$

Therefore, the sign factors of all cases are given by $\text{sgn}(k+l)$ and

$$\frac{\mathbf{v}_l^T \gamma^0 \mathbf{v}_{-k}}{\sqrt{4\omega_k \omega_l}} = \text{sgn}(k+l) \sqrt{\frac{1}{2} + M_{kl}} \tag{179}$$

Using $(\gamma^0)^\dagger = \gamma^0$ the other factors are given as

$$\frac{\mathbf{v}_{-k}^\dagger \gamma^0 \mathbf{v}_l^*}{\sqrt{4\omega_k \omega_l}} = \text{sgn}(k+l) \sqrt{\frac{1}{2} + M_{kl}}. \tag{180}$$

In order to derive the final expression for the interaction eq.(60), it is used that $M_{kl} = M_{lk}$, such that all term are invariant under $k \leftrightarrow l$.

18.2 Charge Conjugated Hamiltonian

It will be shown that the Hamiltonian respects charge conjugation. Starting from section 8.2, it is left to show, that $\sum_{n,m} \text{sgn}(n+m) \sqrt{\frac{1}{2} - M_{nm}} (c_{-n}^\dagger d_m^\dagger + d_n c_{-m})$ conserve charge conjugation. For that, the sum over $\{n, m\}$ is separated into $\{-n < m\}$ and $\{-n > m\}$ and the variables are renamed $\{n, m\} \rightarrow \{-m, -n\}$. $M_{n,m} = M_{-m,-n}$ is invariant in this transformation (see eq.(59)), so

$$\begin{aligned}
\sum_{n,m} \text{sgn}(n+m) d_n c_{-m} &= \sum_{-n < m} \text{sgn}(n+m) d_n c_{-m} + \sum_{-n > m} \text{sgn}(n+m) d_n c_{-m} \\
&= \sum_{-n < m} \text{sgn}(n+m) d_n c_{-m} + \sum_{m > -n} \text{sgn}(-m-n) d_{-m} c_n \\
&\Leftrightarrow \sum_{n,m} \text{sgn}(n+m) d_n c_{-m} = \sum_{-n < m} \text{sgn}(n+m) (d_n c_{-m} + c_n d_{-m})
\end{aligned} \tag{181}$$

$$\rightarrow \sum_{n,m} \text{sgn}(n+m) c_{-n}^\dagger d_m^\dagger = \sum_{-n < m} \text{sgn}(n+m) (c_{-n}^\dagger d_m^\dagger + d_{-n}^\dagger c_m^\dagger). \tag{182}$$

The two terms on the r.h.s. are charge conjugate of each other, so the form is indeed $h = h' + \hat{C} h' \hat{C}$.

Note that (see eq.(59))

$$\sqrt{\frac{1}{2} - M_{-n,n}} = 0,$$

so the cases $\{-n = m\}$ can be excluded.

18.3 Multiplicity Factor for Intermediate States

This chapter shows, that for indistinguishable particles, intermediate states would get a multiplicity factor and certain diagrams can not be separated. By distinguish particles artificially, also different diagrams can be identified and the multiplicity depend only on external particles as expected.

Note that, in a field description like in time ordered perturbation theory, the following calculations wouldn't be needed. But the truncated Hamiltonian is written in a particle description and the effective Hamiltonian is neither local nor Hermitian. Hence, a field description don't work and new Feynman rules must be found.

Bosons:

In general, a field or a boson operator, which is Wick contracted with a state gives an additional multiplicity factor of $\sqrt{n}$ or $\sqrt{n+1}$ according to the number n of bosons in a state (see eq.(28)).

The factor n comes from n indistinguishable particles. But an equivalent description is to distinguish the particles artificially and sum over all possible Wick contractions. For example

$$\phi \frac{1}{\sqrt{n!}} (a_k^\dagger)^n |0\rangle \hat{=} \phi \frac{1}{\sqrt{n!}} \prod_{i=1}^n a_{k,i}^\dagger |0\rangle = \sum_{i=1}^n \phi a_{k,i}^\dagger \left(\frac{1}{\sqrt{n!}} \prod_{j \neq i}^n a_{k,j}^\dagger |0\rangle \right) = \sqrt{n} \phi a_k^\dagger \left(\frac{1}{\sqrt{(n-1)!}} (a_k^\dagger)^{n-1} |0\rangle \right), \quad (183)$$

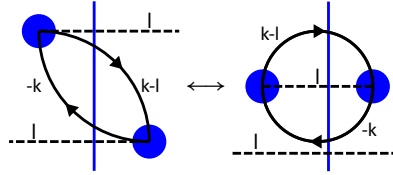
where the first index k is the physical momentum and i distinguish the operators.

All Wick contractions are of that type, so this is always a valid mathematical trick, which yields the same results (although it may seems unphysical).

This step is needed to distinguish different Feynman diagrams with the same intermediate state. Meaning, one term in

$$\tilde{H}_2 = \sum_{\alpha} \frac{\langle f|V|\alpha\rangle \langle \alpha|V|i\rangle}{E_i - E_{\alpha}} \quad (184)$$

with a certain $|\alpha\rangle$ can be associated with multiple Feynman diagrams. For Example



could not be distinguished. The loops are not understood to be summed here. It is one summand with certain momentum k and n , defined by the intermediate state $|\alpha\rangle$. The blue markings highlight, which parts actually contribute to the magnitude. That are the vertices and the intermediate state, but the connections are not defined.

If the external states $|n\rangle$ have n bosons of momenta l , the amplitude of both diagrams together (omitting the factor of $1/(E_i - E_{\alpha})$) is

$$\begin{aligned} \langle n | \overline{\phi} \psi \psi c_{-k}^\dagger d_{k-l}^\dagger | n+1 \rangle \langle n+1 | d_{k-l} c_{-k} \overline{\phi} \psi \psi | n \rangle &= \frac{(n+1)^2 (n!) 2}{n! (n+1)!} \overline{\phi} a_l^\dagger a_l \phi \left(\overline{\psi} \psi c_{-k}^\dagger d_{k-l}^\dagger d_{k-l} c_l \overline{\psi} \psi \right) \\ &= (n+1) \overline{\phi} a_l^\dagger a_l \phi \cdot L \end{aligned} \quad (185)$$

It will be derived, that n times the amplitude corresponds to the diagram, which connects n external particles and the $+1$ comes from the disconnected vacuum loop, which is thereby independent of the n spectator particles.

The first step is to distinguish the particles in $|\alpha\rangle = |n+1\rangle$ (omitting the fermion) by attaching a second index. $\langle\alpha|$ is defined through $\langle\alpha| = (|\alpha\rangle)^\dagger$. Then, the propagator can be identified as

$$\begin{aligned} \frac{1}{(n+1)!} \overline{\phi(a_l^\dagger)^{n+1}} |0\rangle \langle 0| \overline{(a_l)^{n+1}} \phi &= \frac{1}{n+1} \sum_{i=1}^{n+1} \overline{\phi a_{l,i}^\dagger} \overline{a_{l,i}} \phi \frac{1}{n!} \left(\prod_{j \neq i} a_{l,j}^\dagger \right) |0\rangle \langle 0| \left(\prod_{h \neq i} a_{l,h} \right) + \sum_{i \neq j} \overline{\phi a_{l,i}^\dagger} \overline{a_{l,j}} \phi \dots \\ &= 1 \cdot \overline{\phi \phi} |n\rangle \langle n| + \text{other contractions.} \end{aligned} \quad (186)$$

When the fields are contracted to the same auxiliary index, the corresponding diagram have a boson propagator. The leftover boson part $|n\rangle \langle n|$ is normalized and gives one, when contracted with external states.

In the other part, the fields are contracted to different indices, while the associated particles with the same indices are contracted to external states. This configurations are identified with external particles, which are passing by a vertex, like in the first diagram. For the outgoing state, this gives

$$\begin{aligned} &\frac{1}{\sqrt{n!}(n+1)!} \langle 0| (a_l)^n \phi (a_l^\dagger)^{n+1} |0\rangle \langle 0| \overline{(a_l)^{n+1}} \phi \\ &= \frac{1}{\sqrt{n!}(n+1)} \underbrace{\sum_{i=1}^{n+1} \langle 0| \overline{(a_l)^n} a_{l,i}^\dagger \overline{a_{l,i}} \phi}_{(n+1) \cdot n} \left(\phi |n\rangle \langle n| \right) + \frac{1}{\sqrt{n!}} \langle 0| (a_l)^n \overline{\phi \phi} \left(|n\rangle \langle n| \right) \\ &= \sqrt{n} \overline{a_l} \phi \langle n-1| \phi |n\rangle \langle n| + \overline{\phi \phi} \langle n|, \end{aligned} \quad (187)$$

where the definition of the propagator is used. Analogous results gives the in-state, where the residual of the last expression $\phi |n\rangle \langle n|$ is used. Repeating the calculation, but with $|n\rangle$ instead of $|n+1\rangle$ give

$$\frac{1}{\sqrt{n!}n!} \phi |n\rangle \langle n| \cdot |n\rangle = \frac{1}{\sqrt{n!}n} \left(|n-1\rangle \langle n-1| \right) \underbrace{\sum_{i=1}^n \overline{\phi a_{l,i}^\dagger} \overline{a_{l,i}} (a_l^\dagger)^n |0\rangle}_{n \cdot n} = \sqrt{n} \overline{\phi a_l^\dagger} |n-1\rangle. \quad (188)$$

Putting the last two expressions together and adding the fermion loop gives

$$\begin{aligned} \langle n| \phi \overline{\psi} \psi c_{-k}^\dagger d_{k-l}^\dagger |n+1\rangle \langle n+1| d_{k-l} c_{-k} \phi \overline{\psi} \psi |n\rangle &= \sqrt{n^2} \overline{a_l} \phi \phi a_l^\dagger \langle n-1| n-1\rangle \cdot L + \overline{\phi \phi} \langle n| n\rangle \cdot L \\ &= (n+1) \overline{\phi \phi} \cdot L. \end{aligned} \quad (189)$$

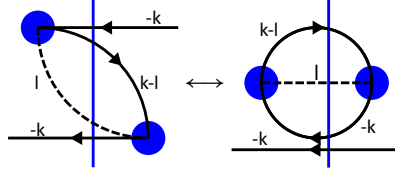
It shows, that by distinguish the particles one recover the Feynman rules, which are independent of the multiplicity of intermediate state. This is important, because otherwise loops would depend on spectator particles.

It is also derived, that certain diagrams can only be distinguished by a purely mathematical trick. But in fact, both diagrams are part of the same term in the old fashioned perturbation theory eq.(184).

Fermions:

The calculations are repeated for fermions. The multiplicity for intermediate states refers to the Pauli exclusion principle here.

For an example the diagrams



are used. In particular the first diagram demonstrate, that every Z-shaped, one particle diagram would be forbidden. But, the corresponding vacuum diagram has two identical fermions as well. It will be shown, that there are always two diagrams, which cancel each other out, even if the Pauli exclusion principle is ignored.

Even with distinguished fermions, the intermediate state $c_{k,1}^\dagger c_{k,2}^\dagger |0\rangle = 0$ doesn't make sense and must vanish, since the numbering is arbitrary, but anti-commuting the operators give a minus sign. On the other hand, the operator $|\alpha\rangle\langle\alpha| = |\alpha\rangle(|\alpha\rangle)^\dagger$ could be taken as not vanishing, since

$$\begin{aligned} c_{k,1}^\dagger c_{k,2}^\dagger |0\rangle\langle 0| c_{k,2} c_{k,1} &= c_{k,2}^\dagger c_{k,1}^\dagger |0\rangle\langle 0| c_{k,1} c_{k,2} \\ c_{k,1}^\dagger c_{k,2}^\dagger |0\rangle\langle 0| c_{k,1} c_{k,2} &= -c_{k,1}^\dagger c_{k,2}^\dagger |0\rangle\langle 0| c_{k,2} c_{k,1}. \end{aligned}$$

Of cause, all these diagrams must still give zero in the end, but at this point, the operator can be kept, since the negative configurations can be identified. The expression $|\alpha\rangle(|\alpha\rangle)^\dagger$ restricts the intermediate operator to take the form of the l.h.s or the r.h.s of the first line. The two possibilities imply a normalization of $1/2$.

In the calculation of the vacuum diagram the propagator is defined in the form of

$$\begin{aligned} \frac{1}{2} \left(\langle 0| \overline{c_{-k} \phi \bar{\psi} \psi} c_{-k,1}^\dagger c_{-k,2}^\dagger |0\rangle \langle 0| \overline{c_{-k,2} c_{-k,1} \phi \bar{\psi} \psi} c_{-k}^\dagger |0\rangle + 1 \leftrightarrow 2 \right) &= \frac{1}{2} \sum_{i=1}^2 \left[\overline{\psi c_{-k,i}^\dagger} c_{-k,i} \overline{\psi} \right] \left(c_{-k} c_{-k}^\dagger \right)^2 .L, \\ &= \langle 1|1\rangle \left[\overline{\psi \psi} \right] .L, \end{aligned} \quad (190)$$

where $|1\rangle$ is a external state containing one of the examined fermion.

The propagator is a matrix and the omitted loop L acts from both sides. The expression is identical to the Z shaped diagram, but the intermediate particles are swapped, which give a minus sign:

$$\begin{aligned} -\frac{1}{2} \left(\langle 0| \overline{c_{-k} \phi \bar{\psi} \psi} c_{-k,2}^\dagger c_{-k,1}^\dagger |0\rangle \langle 0| \overline{c_{-k,2} c_{-k,1} \phi \bar{\psi} \psi} c_{-k}^\dagger |0\rangle + 1 \leftrightarrow 2 \right) &= -\frac{1}{2} \sum_{\substack{i=1 \\ j \neq i}}^2 \left[\overline{\psi c_{-k,j}^\dagger} c_{-k,j} \overline{1} \right] \cdot \left[1 \overline{c_{-k,i}^\dagger} c_{-k,i} \psi \right] .L \\ &= - \left[\overline{\psi} 1 \right] \left[1 \psi \right] .L. \end{aligned} \quad (191)$$

Two identical fermions in the intermediate state can always be anti-commuted in that way. The configurations are associated to different diagrams, but one gets minus sign.

If all diagrams are considered, the Pauli exclusion principle can be ignored for intermediate states, since the diagrams cancel in the end. But, if one diagram is not included, some unphysical interactions might still appear in the theory.

Therefore, all diagrams in a certain order of the coupling must be included to get physical results. This matter is discussed in section 12.